

Modeling and Analysis of Dynamic Systems

Vibration of Two-Degrees-of-Freedom (2DOF) Systems

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Chapter Overview and Learning Objectives

- This Chapter presents the essential concepts in the modeling, analysis, and physics of two-degree-of-freedom (2DOF) vibratory dynamic systems.
- Transition from SDOF to 2DOF systems:
 - Modeling and equations of motion are more complicated
 - Use of matrix algebra in the analysis
 - Develop basic concepts of natural frequencies and modes by solving the eigenvalue problem of undamped system
 - foundation to study multi-degree-of-freedom (MDOF) systems
 - develop techniques for analytical and numerical solutions

Basic Equations:

■ Particle dynamics

$$\sum_{\text{free-body diagram}} \mathbf{F} = \underbrace{\frac{D}{Dt}(m \mathbf{v})}_{\text{Kinematics}} = m \mathbf{a}, \quad \frac{D}{Dt} = \text{time derivative with respect to a stationary frame}$$

$$\Delta T + \Delta V = U_{nc} \quad (\text{non-conservative work})$$

$$\int_{t_1}^{t_2} \sum \mathbf{F} dt = m \mathbf{v}_2 - m \mathbf{v}_1 \triangleq \mathbf{G}_2 - \mathbf{G}_1 \quad \int_{t_1}^{t_2} \sum \mathbf{M}_A dt = (\mathbf{H}_A)_2 - (\mathbf{H}_A)_1, \quad \mathbf{H}_A = \mathbf{r} \times m \mathbf{v}$$

■ Rigid body dynamics

$$\sum_{\text{free-body diagram}} \mathbf{F} = \underbrace{\frac{D}{Dt}(m \mathbf{v}_G)}_{\text{Kinematics}} = m \mathbf{a}_G, \quad G = \text{center of mass}$$

Similar energy and momentum principles with minor modifications

$$\int_{t_1}^{t_2} \sum \mathbf{F} dt = m(\mathbf{v}_G)_1 - m(\mathbf{v}_G)_2$$
$$\sum \mathbf{M}_A = \frac{D}{Dt} \mathbf{H}_G + \mathbf{R}_G \times m \mathbf{a}_G$$

1. **Conservative Systems (No Nonconservative Forces/No Energy Dissipation):**

Kinetic energy = T

Potential energy = V

Conservation of energy: $T + V = \text{const}$

Differentiate with respect to time t

2. **Lagrange's Equations:**

Lagrangian $L = T - V$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = Q_i \quad \text{for } i = 1, 2, \dots, n$$

n = number of degrees of freedom

Q_i = generalized force corresponding to generalized coordinate q_i .

Find Q_i using: $\delta W = \sum Q_i \delta q_i$

where δW = work done by nonconservative forces in a general incremental motion $(\delta q_1, \delta q_2, \dots, \delta q_n)$.

3. **Newtonian Approach:**

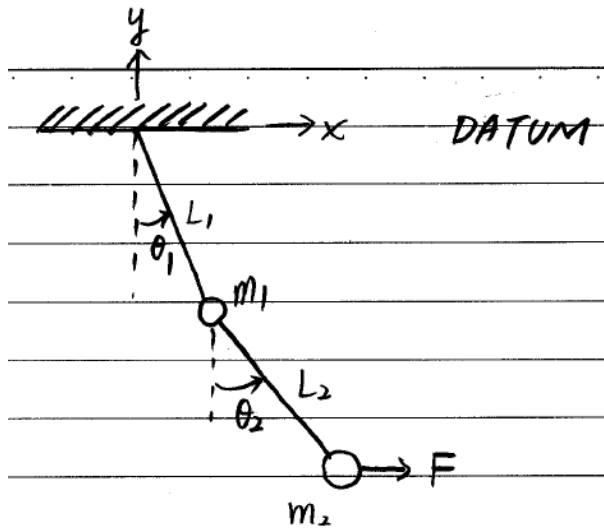
$$\sum \text{Forces} = \frac{d}{dt} \sum \text{Linear momentum}$$

$$\sum \text{Torques} = \frac{d}{dt} \sum \text{Angular momentum} \quad (\text{About centroid or a fixed point})$$

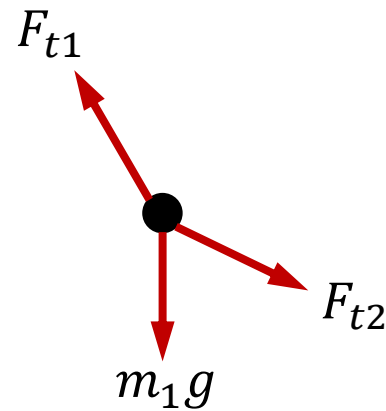
Brief Discussion on Energy-Based Lagrange's Equations of Motion

Shortcomings of Newtonian Approach

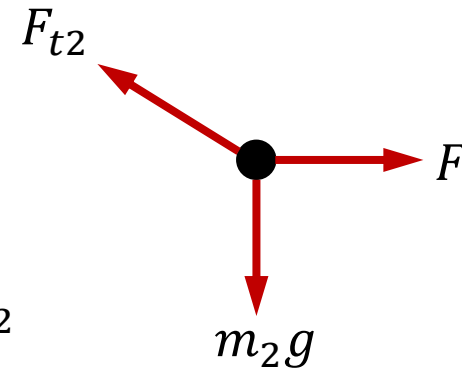
- Vector Calculus based approach may result in unnecessary equations and requires free-body diagrams which include internal forces.
- Derivation of governing equations of motion requires consideration of internal forces, rendering the problem formulation complicated. Moreover, these internal forces are often not of interest.
- In general, not easily amenable to algorithmic numerical coding.



FBD for m_1



FBD for m_2



The internal forces (tension forces) F_{t1} and F_{t2} are often not of interest. The system has 2 degrees of freedom but Newtonian approach leads to 4 equations (for 4 unknowns θ_1 , θ_2 , F_{t1} , and F_{t2}). Thus, two unnecessary equations.

Hamilton's Principle

The extended Hamilton's Principle states:

$$\int_{t_1}^{t_2} (\delta T + \delta W) dt = \delta \int_{t_1}^{t_2} (T + W) dt = 0 \quad (8)$$

where δ represents a virtual displacement (infinitesimal displacement without change in time). The above states that the motion between t_1 and t_2 is such that the energy functional

$$\int_{t_1}^{t_2} (T + W) dt$$

is stationary (minimum) for the actual path compared to neighbouring paths between the same two end points in time. This is the fundamental variational problem in mechanics (cf. Calculus of Variations).

Lagrange's equations of motion

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For a system of n particles, express the displacement vectors as a function of the generalized coordinates:

$$\mathbf{r}_i = \mathbf{r}_i(q_1, q_2, \dots, q_N, t), \quad i = 1, 2, \dots, n \quad (1)$$

The kinetic energy of the system may be expressed as:

$$T = \sum_{i=1}^n \frac{1}{2} m_i \dot{\mathbf{r}}_i \cdot \dot{\mathbf{r}}_i \quad \text{where, } \dot{\mathbf{r}}_i = \sum_{j=1}^N \frac{\partial \mathbf{r}_i}{\partial q_j} \dot{q}_j + \frac{\partial \mathbf{r}_i}{\partial t} \quad (2)$$

Thus, the kinetic energy can be expressed as:

$$T = T(q_1, q_2, \dots, q_N, \dot{q}_1, \dot{q}_2, \dots, \dot{q}_N, t) \quad (3)$$

The variation of the kinetic energy term is:

$$\delta T = \sum_{j=1}^N \frac{\partial T}{\partial q_j} \delta q_j + \frac{\partial T}{\partial \dot{q}_j} \delta \dot{q}_j, \quad \text{where } \delta \dot{q}_j = \frac{d}{dt}(\delta q_j) \quad (4)$$

Note: no time change in the variation operator δ

Lagrange's equations of motion

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Based on the definition of work done by a force, the virtual work done is:

$$\delta W = \sum_{k=1}^K \mathbf{F}_k \cdot \delta \mathbf{r}_k, \text{ where there are } K \text{ force vectors} \quad (5)$$

Thus,

$$\delta W = \sum_{k=1}^K \sum_{j=1}^N \mathbf{F}_k \cdot \frac{\partial \mathbf{r}_k}{\partial q_j} \delta q_j = \sum_{j=1}^N Q_j \delta q_j, \quad Q_j = \sum_{k=1}^K \mathbf{F}_k \cdot \frac{\partial \mathbf{r}_k}{\partial q_j} \quad (6)$$

Note that Q_j are the generalised forces and are derived from the virtual work done. In practice, evaluate Eq. (5) and then use the relation of Eq. (1) to obtain Q_j . Substituting all these terms into the extended Hamilton's Principle gives:

$$\int_{t_1}^{t_2} (\delta T + \delta W) dt = \int_{t_1}^{t_2} \left(\sum_{j=1}^N \frac{\partial T}{\partial q_j} \delta q_j + \frac{\partial T}{\partial \dot{q}_j} \delta \dot{q}_j \right) dt + \int_{t_1}^{t_2} \sum_{j=1}^N Q_j \delta q_j dt \quad (7)$$

Lagrange's equations of motion

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To simplify the above expression, we write the second term on the RHS of Eq. (7) in terms of δq_j . Apply integration by parts technique and $\delta q_j = 0$ at t_1 and t_2 :

$$\int_{t_1}^{t_2} \sum_{j=1}^N \frac{\partial T}{\partial \dot{q}_j} \delta \dot{q}_j dt = \int_{t_1}^{t_2} \sum_{j=1}^N -\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) \delta q_j dt + \sum_{j=1}^N \frac{\partial T}{\partial \dot{q}_j} \delta q_j \Big|_{t_1}^{t_2} \quad (8)$$

Substituting Eq. (8) into Eq. (7) leads to the Euler-Lagrange equation (similar to the procedure that we have learned in the subject of Calculus of Variations)

$$\int_{t_1}^{t_2} (\delta T + \delta W) dt = - \sum_{j=1}^N \int_{t_1}^{t_2} \left[\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} - Q_j \right] \delta q_j dt = 0 \quad (9)$$

Since the virtual displacements δq_j and the time limits are arbitrary, the integrand must vanish, leading to:

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} - Q_j = 0 \quad (10)$$

Lagrange's equations of motion

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Thus, we obtain the Lagrange's equation of motion:

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = Q_j, \quad j = 1, 2, \dots, N \quad (10)$$

If one writes the virtual work done in terms of the conservative and non-conservative parts, $\delta W = \delta W_c + \delta W_{nc}$, where $\delta W_c = -\delta V$, then we obtain an alternative form of the Lagrange's equations of motion in terms of the Lagrange's function, $L = T - V$, assuming that V is not a function of the generalised velocity $\dot{q}_j$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} = (Q_j)_{nc}, \quad j = 1, 2, \dots, N \quad (11)$$

Notes: The Lagrange's equations are derived from an energy-based approach, requiring only the evaluation of the energy terms and the generalized forces. This approach significantly simplifies the procedure to obtain the governing equations of motion. Moreover, the approach is amenable to algorithmic numerical coding. Equation (11) is generally easier to use than Eq. (10) since one only needs to evaluate the non-conservative generalized forces.

Lagrange's equations of motion with damping

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Damping can be represented by a dissipation function $\mathcal{D}$. For linear systems, $\mathcal{D}$ is usually expressed in terms of the generalized velocities:

$$\mathcal{D} = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n c_{ij} \dot{q}_i \dot{q}_j \quad (12)$$

The Lagrange's equations are then modified as

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} + \frac{\partial \mathcal{D}}{\partial \dot{q}_j} = (Q_j)_{nc}, \quad j = 1, 2, \dots, N \quad (13)$$

Note that some textbooks define the dissipation function as:

$$\mathcal{D} = -\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n c_{ij} \dot{q}_i \dot{q}_j \quad (14)$$

to keep with the connotation of “dissipation” (energy lost). Using that definition, the equations are:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} = (Q_j)_{nc} + \frac{\partial \mathcal{D}}{\partial \dot{q}_j}, \quad j = 1, 2, \dots, N \quad (15)$$

Lagrange's Equations: Simple pendulum example

Consider the simple pendulum problem. Assume that the connecting cable is massless and taut. Derive the governing equation of motion. Choose $\theta(t)$ as the generalized coordinate.

Using kinematics, $\vec{v} = l \dot{\theta} \hat{e}_t$ ($\hat{e}_t$ = tangential directional vector, as shown)

$$\Rightarrow T = \frac{1}{2} m v^2 = \frac{1}{2} l^2 \dot{\theta}^2$$

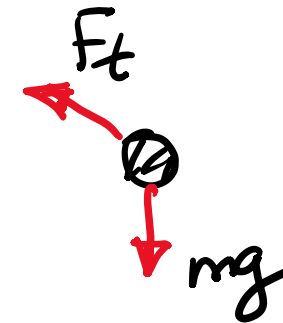
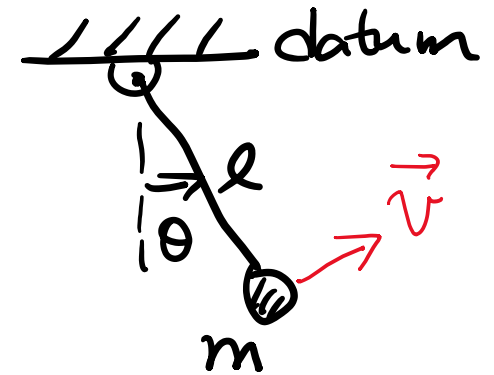
$$V = mg(-l \cos \theta)$$

$$\Rightarrow L = \frac{m}{2} l^2 \dot{\theta}^2 + mgl \cos \theta$$

Conservative system (understand why)

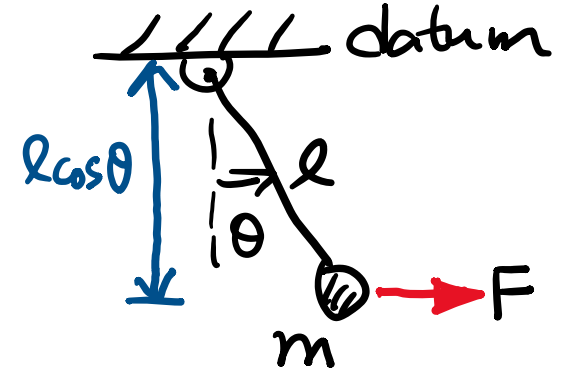
$$\Rightarrow Q_{nc} = 0$$

$$\text{Lagrange's eqn } \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = 0 \Rightarrow ml^2 \ddot{\theta} + mgl \sin \theta = 0$$



Lagrange's Equations: Simple pendulum example

Consider the simple pendulum problem again. Suppose a horizontal force is acting on the pendulum. Derive the governing equation of motion and explain the *physical meaning* of the generalized force for this problem. Again, choose $\theta(t)$ as the generalized coordinate.



The kinetic and potential energies are still the same. But the generalised force is not zero. To determine the generalised force, determine the virtual work done by the applied force.

$$\delta W_{nc} = F \mathbf{i} \cdot \delta \mathbf{r}$$

$$\mathbf{r} = \ell \sin \theta \mathbf{i} - \ell \cos \theta \mathbf{j}, \quad \delta \mathbf{r} = \ell \cos \theta \delta \theta \mathbf{i} + \ell \sin \theta \delta \theta \mathbf{j}$$

$$\delta W_{nc} = F \mathbf{i} \cdot \delta \mathbf{r} = F \ell \cos \theta \delta \theta. \text{ Hence, } Q_\theta = F \ell \cos \theta$$

The governing equation of motion is now:

$$m \ell^2 \ddot{\theta} + m g \ell \sin \theta = F \ell \cos \theta$$

The generalised force is the moment of the applied force about the pivot point.

Modeling and Free Vibration of 2DOF Systems

Example: A string with two masses

The system consists of 2 masses suspended on a massless string of tension T and subjected to external force $F_1(t)$ and $F_2(t)$, F_T is tension in string.

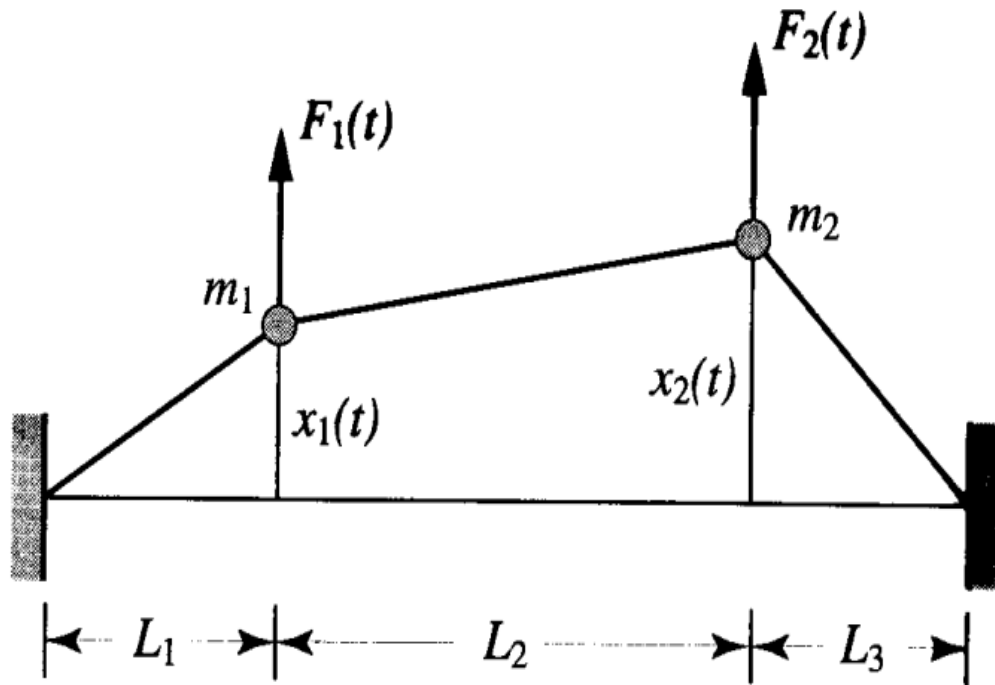


Fig 1.a. Two masses on a string

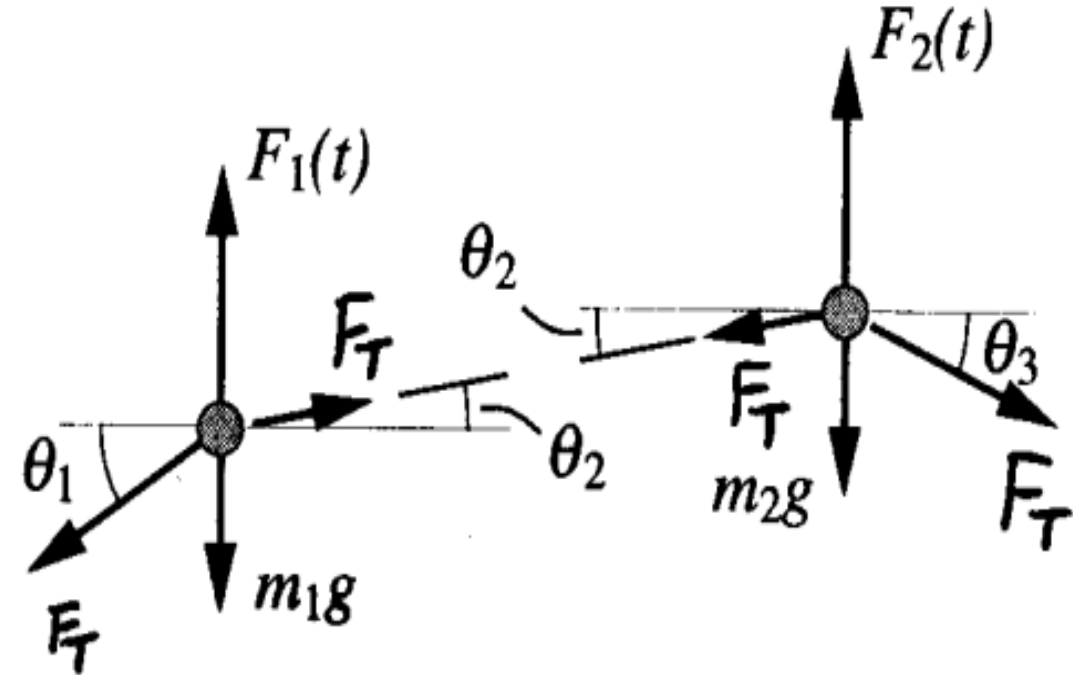


Fig 1.b. Free-body-diagrams

Solution: A string with two masses

The Newton's equations of motion are

$$m_1 \ddot{x}_1 = F_1 - m_1 g - F_T \sin \theta_1 + F_T \sin \theta_2$$

$$m_2 \ddot{x}_2 = F_2 - m_2 g - F_T \sin \theta_2 + F_T \sin \theta_3$$

By assuming that the tension is uniform, the problem is meaningful only for small displacement. Otherwise, tension cannot be maintained constant.

$$\text{Thus, } \sin(\theta_1) \approx \frac{x_1}{L_1}, \quad \sin(\theta_2) \approx \frac{x_2 - x_1}{L_2}, \quad \sin(\theta_3) \approx \frac{x_2}{L_3}$$

Also, let $\bar{x}_1 = x_1 - x_{e1}$, $\bar{x}_2 = x_2 - x_{e2}$, where the notations x_{e1} and x_{e2} represent the equilibrium positions under weight

$$\begin{cases} m_1 \ddot{\bar{x}}_1 + \left(\frac{F_T}{L_1} + \frac{F_T}{L_2} \right) \bar{x}_1 - \frac{F_T}{L_2} \bar{x}_2 = F_1 \\ m_2 \ddot{\bar{x}}_2 + \left(\frac{F_T}{L_2} + \frac{F_T}{L_3} \right) \bar{x}_2 - \frac{F_T}{L_2} \bar{x}_1 = F_2 \end{cases}$$

Solution: A string with two masses

Lagrange's equations of motion

$$\delta W_{ne} = F_1 \delta x_1 + F_2 \delta x_2 = -F_T \sin \theta_1 \delta x_1 + F_T \sin \theta_2 \delta x_1 - F_T \sin \theta_2 \delta x_2 - F_T \sin \theta_3 \delta x_2$$

$$T = \frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_2^2, \quad V = m_1 g x_1 + m_2 g x_2$$

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_k} \right) - \frac{\partial T}{\partial q_k} + \frac{\partial V}{\partial q_k} = (Q_k)_{nc}, \quad k = 1, 2$$

$$m_1 \ddot{x}_1 + m_1 g = F_1 - F_T \sin \theta_1 + F_T \sin \theta_2$$

$$m_2 \ddot{x}_2 + m_2 g = F_2 - F_T \sin \theta_2 - F_T \sin \theta_3$$

where, $\theta_1, \theta_2, \theta_3$ are assumed small.

Example: A 2DOF bar

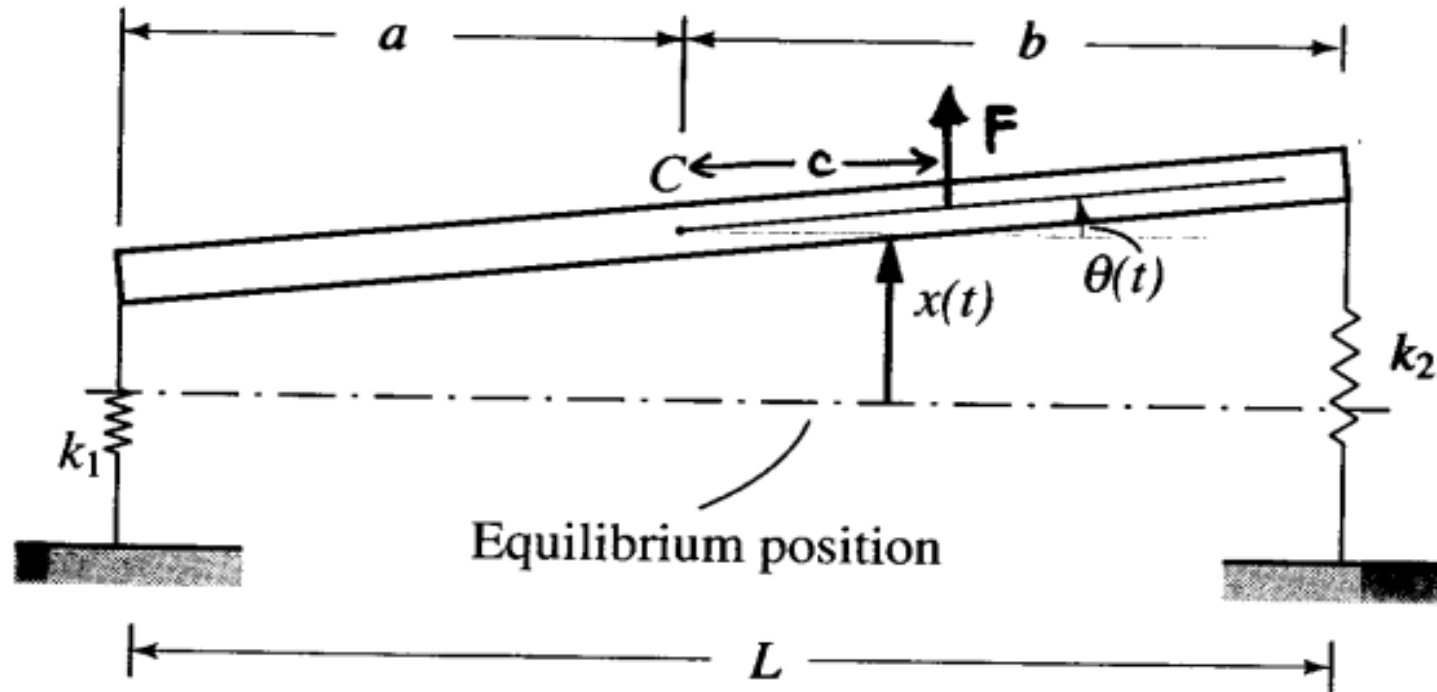


Fig.2 Simplified model of an automobile

Point C is the location of the cm. “ mg ” not considered since the vibration measurement is made from the equilibrium. Assuming small oscillations, the governing equations of motion are:

$$\begin{aligned} m\ddot{x} + (k_1 + k_2)x - (k_1a - k_2b)\theta &= F \\ I_c\ddot{\theta} - (k_1a - k_2b)x + (k_1a^2 + k_2b^2)\theta &= Fc \end{aligned}$$

Example: A 2DOF spring-mass system

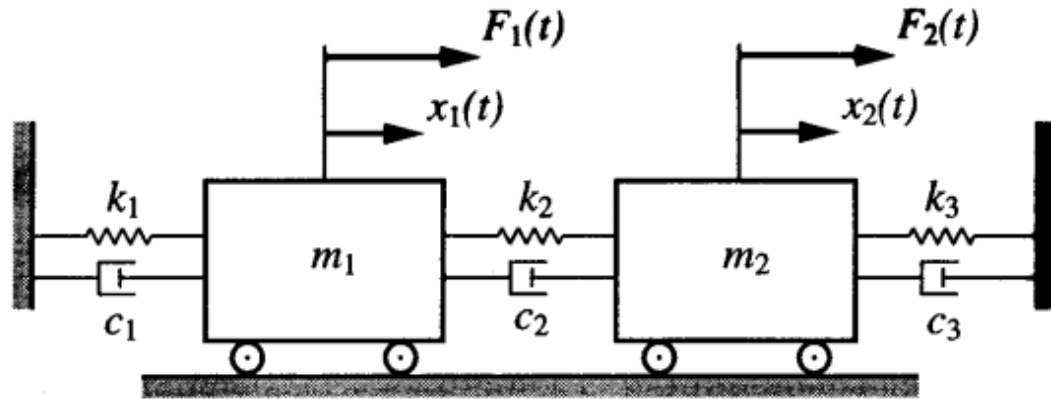


Fig.3 (a) Damped two-degree-of-freedom system in horizontal vibration

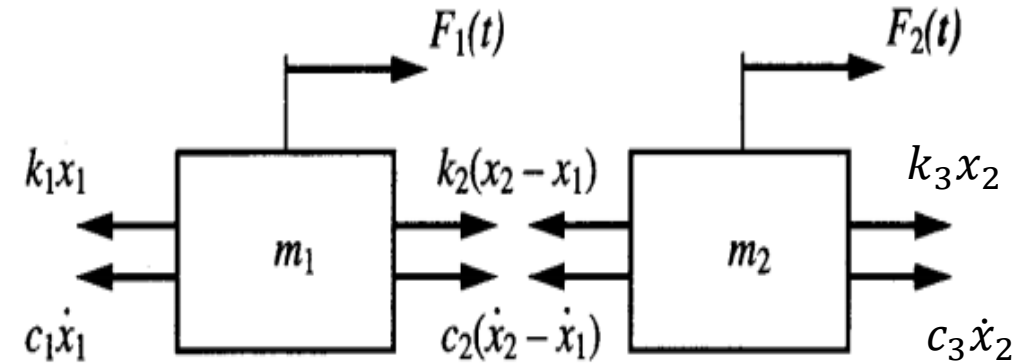


Fig. 3(b) Free-body diagrams

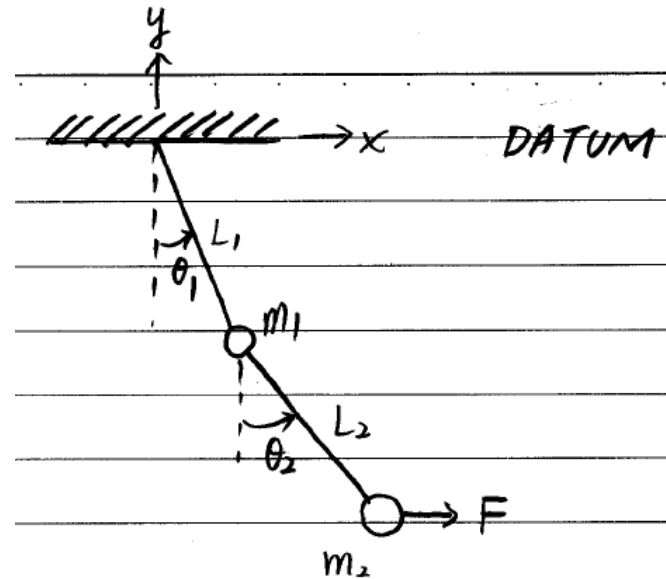
Solution:

The governing equations in the form of matrix:

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{pmatrix} + \begin{bmatrix} c_1 + c_2 & -c_2 \\ -c_2 & c_2 + c_3 \end{bmatrix} \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 + k_3 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} F_1 \\ F_2 \end{pmatrix}$$

Lagrange's Equations: Double pendulum example

Consider the double pendulum problem. Assume that the connecting cables are massless and taut. Derive the governing equations of motion for the two generalized coordinates θ_1 and θ_2 under the action of a horizontal force $F(t)$ applied at m_2 .



Lagrange's Equations: Double pendulum example

We have already chosen the generalized coordinates $q_1 = \theta_1$ and $q_2 = \theta_2$. The next step is to find the kinetic energy and potential energy of this system.

$$T = \sum_{i=1}^2 m_i \dot{\mathbf{r}}_i \cdot \dot{\mathbf{r}}_i = \sum_{i=1}^2 m_i \mathbf{v}_i \cdot \mathbf{v}_i = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

Use the Cartesian coordinate system shown, we have

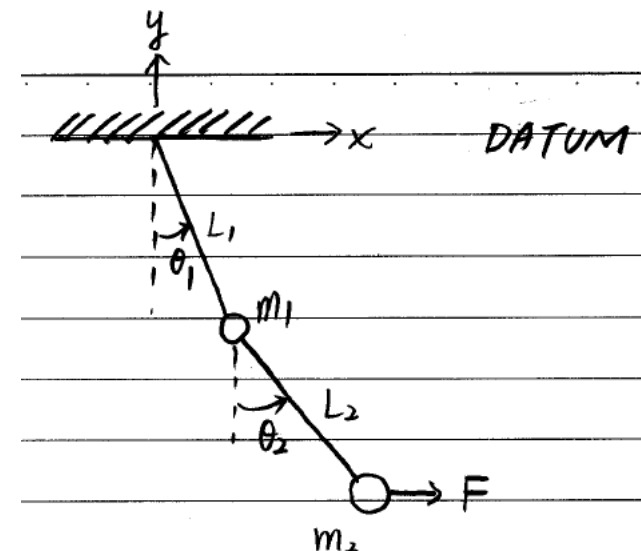
$$\mathbf{r}_1 = L_1 \sin \theta_1 \mathbf{i} - L_1 \cos \theta_1 \mathbf{j}. \text{ Hence, } \mathbf{v}_1 = L_1 \dot{\theta}_1 \cos \theta_1 \mathbf{i} + L_1 \dot{\theta}_1 \sin \theta_1 \mathbf{j} \Rightarrow v_1^2 = L_1^2 \dot{\theta}_1^2$$

$$\mathbf{r}_2 = (L_1 \sin \theta_1 + L_2 \sin \theta_2) \mathbf{i} - (L_1 \cos \theta_1 + L_2 \cos \theta_2) \mathbf{j}. \text{ Hence,}$$

$$\mathbf{v}_2 = (L_1 \dot{\theta}_1 \cos \theta_1 + L_2 \dot{\theta}_2 \cos \theta_2) \mathbf{i} + (L_1 \dot{\theta}_1 \sin \theta_1 + L_2 \dot{\theta}_2 \sin \theta_2) \mathbf{j}$$

$$\Rightarrow v_2^2 = L_1^2 \dot{\theta}_1^2 + L_2^2 \dot{\theta}_2^2 + 2L_1 L_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_2 - \theta_1)$$

$$\text{The potential energy: } V = m_1 g y_1 + m_2 g y_2 = m_1 g (-L_1 \cos \theta_1) + m_2 g (-L_1 \cos \theta_1 - L_2 \cos \theta_2)$$



Lagrange's Equations: Double pendulum example

The Lagrangian function of this system is:

$$\begin{aligned} L &= T - V \\ &= \frac{1}{2} m_1 L_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 [L_1^2 \dot{\theta}_1^2 + L_2^2 \dot{\theta}_2^2 + 2L_1 L_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_2 - \theta_1)] \\ &\quad + m_1 g L_1 \cos \theta_1 + m_2 g (L_1 \cos \theta_1 + L_2 \cos \theta_2) \end{aligned}$$

The virtual work done by the non-conservative force $F(t)$ is:

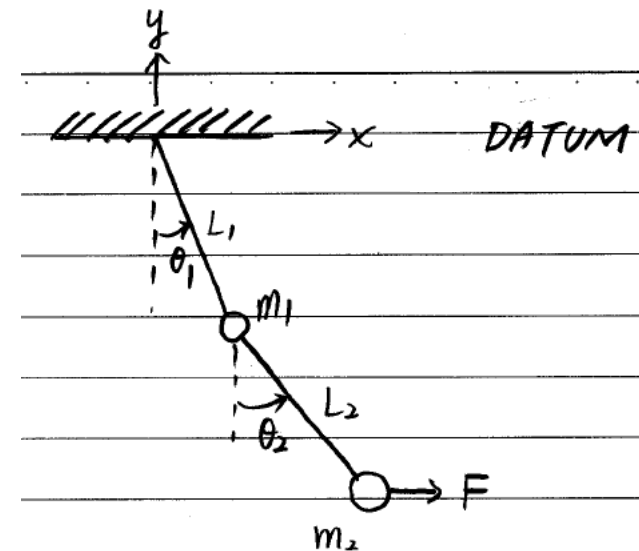
$$\delta W_{nc} = \mathbf{F} \cdot \delta \mathbf{r}_2, \text{ where } \mathbf{F} = F(t) \mathbf{i}$$

Since $\mathbf{r}_2 = (L_1 \sin \theta_1 + L_2 \sin \theta_2) \mathbf{i} - (L_1 \cos \theta_1 + L_2 \cos \theta_2) \mathbf{j}$, thus,

$$\delta \mathbf{r}_2 = (L_1 \cos \theta_1 \delta \theta_1 + L_2 \cos \theta_2 \delta \theta_2) \mathbf{i} + (L_1 \sin \theta_1 \delta \theta_1 + L_2 \sin \theta_2 \delta \theta_2) \mathbf{j}$$

$$\delta W_{nc} = F \mathbf{i} \cdot \delta \mathbf{r}_2 = F(L_1 \cos \theta_1 \delta \theta_1 + L_2 \cos \theta_2 \delta \theta_2)$$

$$\Rightarrow Q_{1nc} = FL_1 \cos \theta_1, Q_{2nc} = FL_2 \cos \theta_2$$

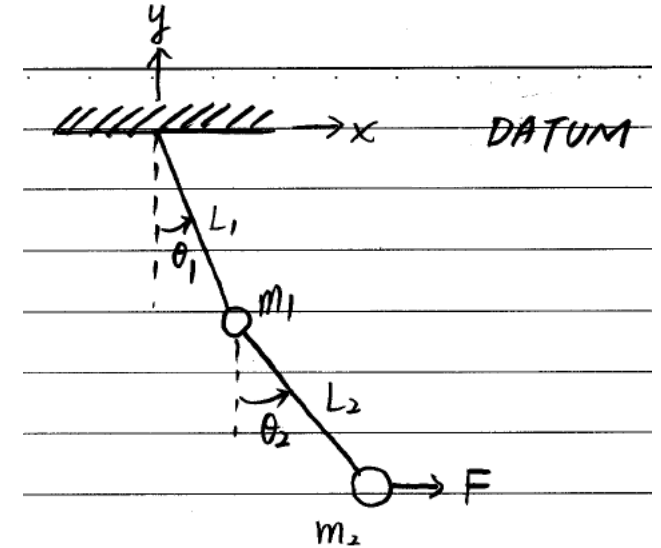


Lagrange's Equations: Double pendulum example

The governing equation of motion for $\theta_1(t)$ is:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_1} \right) - \frac{\partial L}{\partial \theta_1} = Q_{1nc}$$

$$L = \frac{1}{2} m_1 L_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 [L_1^2 \dot{\theta}_1^2 + L_2^2 \dot{\theta}_2^2 + 2L_1 L_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_2 - \theta_1)] + m_1 g L_1 \cos \theta_1 + m_2 g (L_1 \cos \theta_1 + L_2 \cos \theta_2)$$



$$\frac{\partial L}{\partial \dot{\theta}_1} = m_1 L_1^2 \dot{\theta}_1 + m_2 L_1^2 \dot{\theta}_1 + m_2 L_1 L_2 \dot{\theta}_2 \cos(\theta_2 - \theta_1)$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_1} \right) = (m_1 + m_2) L_1^2 \ddot{\theta}_1 + m_2 L_1 L_2 \ddot{\theta}_2 \cos(\theta_2 - \theta_1) - m_2 L_1 L_2 \dot{\theta}_2 \sin(\theta_2 - \theta_1) (\dot{\theta}_2 - \dot{\theta}_1)$$

$$\frac{\partial L}{\partial \theta_1} = m_2 L_1 L_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_2 - \theta_1) - m_1 g L_1 \sin \theta_1 - m_2 g L_1 \sin \theta_1$$

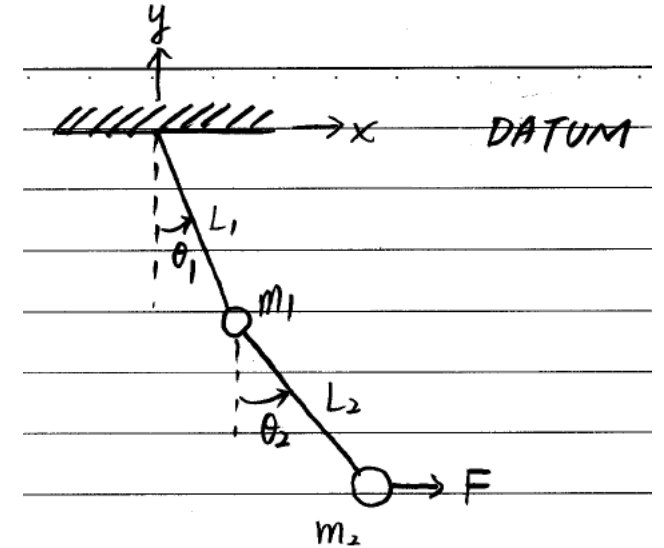
$$\Rightarrow (m_1 + m_2) L_1^2 \ddot{\theta}_1 + m_2 L_1 L_2 \ddot{\theta}_2 \cos(\theta_2 - \theta_1) - m_2 L_1 L_2 \dot{\theta}_2^2 \sin(\theta_2 - \theta_1) + (m_1 + m_2) g L_1 \sin \theta_1 = F L_1 \cos \theta_1$$

Lagrange's Equations: Double pendulum example

The governing equation of motion for $\theta_2(t)$ is:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_2} \right) - \frac{\partial L}{\partial \theta_2} = Q_{2nc}$$

$$L = \frac{1}{2} m_1 L_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 [L_1^2 \dot{\theta}_1^2 + L_2^2 \dot{\theta}_2^2 + 2L_1 L_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_2 - \theta_1)] \\ + m_1 g L_1 \cos \theta_1 + m_2 g (L_1 \cos \theta_1 + L_2 \cos \theta_2)$$



$$\frac{\partial L}{\partial \dot{\theta}_2} = m_2 L_2^2 \dot{\theta}_2 + m_2 L_1 L_2 \dot{\theta}_1 \cos(\theta_2 - \theta_1)$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_2} \right) = m_2 L_2^2 \ddot{\theta}_2 + m_2 L_1 L_2 \ddot{\theta}_1 \cos(\theta_2 - \theta_1) - m_2 L_1 L_2 \dot{\theta}_1 (\dot{\theta}_2 - \dot{\theta}_1) \sin(\theta_2 - \theta_1)$$

$$\frac{\partial L}{\partial \theta_2} = -m_2 L_1 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_2 - \theta_1) - m_2 g L_2 \sin \theta_2$$

$$\Rightarrow m_2 L_2^2 \ddot{\theta}_2 + m_2 L_1 L_2 \ddot{\theta}_1 \cos(\theta_2 - \theta_1) + m_2 L_1 L_2 \dot{\theta}_1^2 \sin(\theta_2 - \theta_1) \\ + m_2 g L_2 \sin \theta_2 = F L_2 \cos \theta_2 \quad ||$$

Lagrange's Equations: Double pendulum example

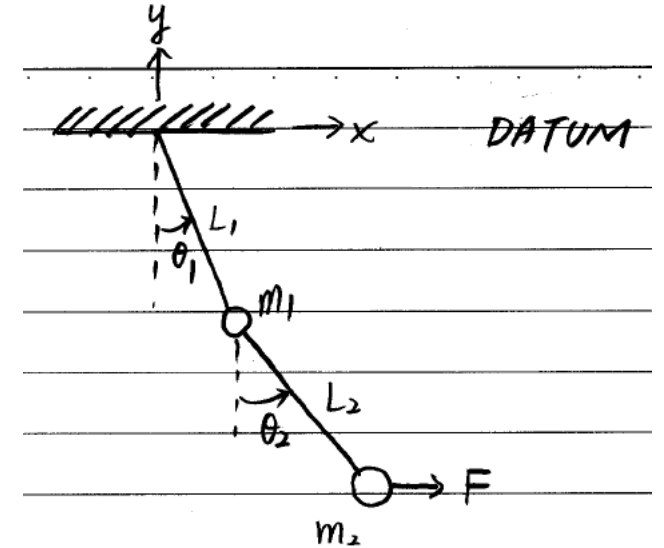
For small angles, the governing equation of motion can be linearized to study the linear vibration problem. The angles $\sin \theta \approx \theta$ and $\cos \theta \approx 1$, and we shall neglect all the nonlinear terms.

$$(m_1 + m_2)L_1^2\ddot{\theta}_1 + m_2L_1L_2\ddot{\theta}_2 + (m_1 + m_2)gL_1\theta_1 = FL_1$$
$$m_2L_2^2\ddot{\theta}_2 + m_2L_1L_2\ddot{\theta}_1 + m_2gL_2\theta_2 = FL_2$$

In matrix form, we have

$$\underbrace{\begin{bmatrix} (m_1 + m_2)L_1^2 & m_2L_1L_2 \\ m_2L_1L_2 & m_2L_2^2 \end{bmatrix}}_{\text{mass matrix } \mathbf{M}} \begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix} + \underbrace{\begin{bmatrix} (m_1 + m_2)gL_1 & 0 \\ 0 & m_2gL_2 \end{bmatrix}}_{\text{stiffness matrix } \mathbf{K}} \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = \begin{Bmatrix} FL_1 \\ FL_2 \end{Bmatrix}$$

Note that the system is coupled through the inertia. With the mass and stiffness matrices, invoke the “eig” function in MATLAB (or by hand calculations) to find the natural frequencies and the mode shapes.



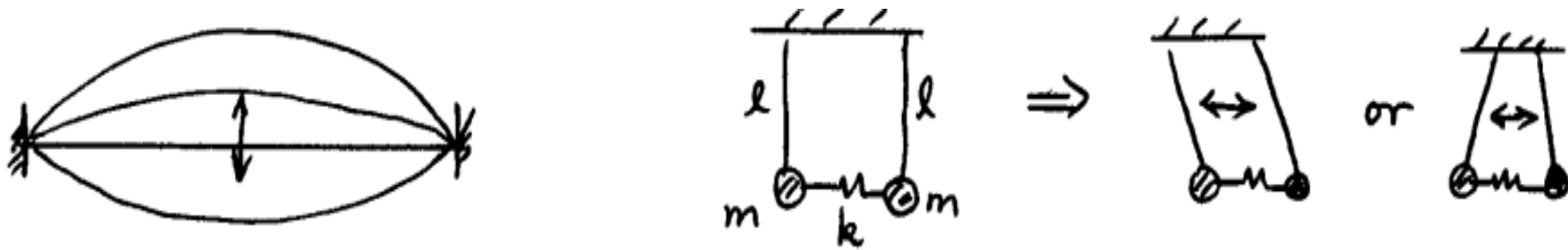
1. Free vibration of undamped systems

Consider the free vibration of a vibratory system of the general form,

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = 0$$

where $\mathbf{M}$, $\mathbf{K}$ are positive-definite, real symmetric matrices .

Fundamental to the study of vibrations are certain special solutions in which the whole system execute the same motion in time (synchronous motions). These are the mode shapes, as illustrated below.



To find these mode shapes (natural modes), we write

$$\mathbf{x}(t) = f(t)\mathbf{u}$$

**Method of separation or
decomposition technique**

where $f(t)$ is a function of the temporal variable, and $\mathbf{u}$ is a function of the spatial variable representing the mode shape. We then, obtain,

$$\mathbf{M}\mathbf{u}\ddot{f} + \mathbf{K}\mathbf{u}f = 0, \quad \Rightarrow \quad \mathbf{M}\mathbf{u}\frac{\ddot{f}}{f} = -\mathbf{K}\mathbf{u}$$

The RHS of the above equation is independent of t , hence we must have, $\ddot{f} + \lambda f(t) = 0$, where the eigenvalue λ is a constant to be determined. We thus establish the standard algebraic eigenvalue problem

$$\mathbf{K}\mathbf{u} = \lambda\mathbf{M}\mathbf{u}, \quad \frac{\mathbf{u}^T \mathbf{K}\mathbf{u}}{\mathbf{u}^T \mathbf{M}\mathbf{u}} = \lambda \quad \text{Algebraic eigenvalue problem}$$

Here, $\mathbf{u}$ is an eigenvector, and λ is the eigenvalue, $\lambda = \omega^2$ (ω is the frequency). The eigenvalues are related to the natural frequencies. Since $(\mathbf{K} - \lambda\mathbf{M})\mathbf{u} = 0$, for non-trivial solutions, we require $\det(\mathbf{K} - \lambda\mathbf{M}) = 0$. This is the frequency equation or characteristic equation whose solution gives the eigenvalues λ . For 2DOF systems, two roots, λ_1, λ_2 are obtained from the frequency equation.

Properties of the Eigenvalue Problem:

1) If $\mathbf{M}$, $\mathbf{K}$ are positive-definite , then

$$\lambda = \frac{\mathbf{u}^T \mathbf{K} \mathbf{u}}{\mathbf{u}^T \mathbf{M} \mathbf{u}} > 0 \quad (\lambda \text{ is positive})$$

2) Since $\mathbf{K} \mathbf{u} = \lambda \mathbf{M} \mathbf{u}$, $\mathbf{u}^T \mathbf{K} \mathbf{u} = \lambda \mathbf{u}^T \mathbf{M} \mathbf{u}$

If $\mathbf{K}$, $\mathbf{M}$ are real and symmetric, then denote λ^* and $\mathbf{u}^*$ as the complex conjugate of λ and $\mathbf{u}$, respectively.

$$\begin{aligned} \mathbf{K} \mathbf{u} &= \lambda \mathbf{M} \mathbf{u} & \mathbf{u}^{*T} \mathbf{K} \mathbf{u} &= \lambda \mathbf{u}^{*T} \mathbf{M} \mathbf{u} \\ \mathbf{K}^* \mathbf{u}^* &= \lambda^* \mathbf{M}^* \mathbf{u}^* & \mathbf{u}^T \mathbf{K}^* \mathbf{u}^* &= \lambda^* \mathbf{u}^T \mathbf{M}^* \mathbf{u}^* \\ & & \mathbf{u}^{*T} \mathbf{K}^{*T} \mathbf{u} &= \lambda^* \mathbf{u}^{*T} \mathbf{M}^{*T} \mathbf{u} \end{aligned}$$

where, $\mathbf{K}^*$ is the adjoint matrix of $\mathbf{K}$, and $\mathbf{M}^*$ is the adjoint matrix of $\mathbf{M}$.

$$\mathbf{u}^{*T} (\mathbf{K} - \mathbf{K}^{*T}) \mathbf{u} = \mathbf{u}^{*T} (\lambda \mathbf{M} - \lambda^* \mathbf{M}^{*T})$$

But $\mathbf{K} = \mathbf{K}^*$ and $\mathbf{M} = \mathbf{M}^*$ $0 = \mathbf{u}^{*T} (\lambda - \lambda^*) \mathbf{M} \mathbf{u}$ $\lambda - \lambda^* = 0$ (thus λ is real)

Thus, λ are real and positive for symmetric eigenvalue problems. Also, λ cannot be zero for positive-definite systems.

3) Eigenvectors $\mathbf{u}_j$ are also real.

Proof: Consider $\mathbf{K}\mathbf{u}_j = \lambda_j \mathbf{M}\mathbf{u}_j$

Since $\mathbf{M}$, $\mathbf{K}$ and λ_j are all real, hence u_j must be real, since complex numbers would never appear in the formulation.

$$\begin{aligned} 4) \quad \mathbf{K}\mathbf{u}_i &= \lambda_i \mathbf{M}\mathbf{u}_i & \text{thus we have, } \mathbf{u}_j^T \mathbf{K} \mathbf{u}_i &= \lambda_i \mathbf{u}_j^T \mathbf{M} \mathbf{u}_i \\ \mathbf{K}\mathbf{u}_j &= \lambda_j \mathbf{M}\mathbf{u}_j & \text{thus we have, } \mathbf{u}_i^T \mathbf{K} \mathbf{u}_j &= \lambda_j \mathbf{u}_i^T \mathbf{M} \mathbf{u}_j \end{aligned}$$

Take the transpose of the second equation

$$\mathbf{u}_j^T \mathbf{K}^T \mathbf{u}_i = \lambda_j \mathbf{u}_j^T \mathbf{M}^T \mathbf{u}_i$$

and subtract from the first equation. Since we have $\mathbf{u}_j^T \mathbf{K} \mathbf{u}_i = \mathbf{u}_j^T \mathbf{K}^T \mathbf{u}_i$ and $\mathbf{u}_j^T \mathbf{M} \mathbf{u}_i = \mathbf{u}_j^T \mathbf{M}^T \mathbf{u}_i$ and thus we obtain

$$0 = (\lambda_i - \lambda_j) \mathbf{u}_j^T \mathbf{M} \mathbf{u}_i$$

where, $\mathbf{u}_i$ and $\mathbf{u}_j$ are mutually orthogonal for $\lambda_i \neq \lambda_j$.

One can also show that $\mathbf{u}_j^T \mathbf{K} \mathbf{u}_i = 0$ and $\mathbf{u}_j^T \mathbf{M} \mathbf{u}_i = 0$ for $\lambda_i \neq \lambda_j$. We say that the eigenvectors or natural or normal modes are mutually orthogonal to each other with respect to the mass matrix.

Free Response solution:

To evaluate the free vibration response (due to initial conditions), consider

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = 0, \quad \mathbf{x}(0) = \begin{pmatrix} x_{10} \\ x_{20} \end{pmatrix}, \quad \dot{\mathbf{x}}(0) = \begin{pmatrix} v_{10} \\ v_{20} \end{pmatrix}$$

The key step in the solution analysis is the **modal analysis**. In general, we can write the response in the general form as:

$$\mathbf{x}(t) = \sum_{k=1}^n q_k(t) \mathbf{u}_k$$

where, q_k is the natural coordinate or principal coordinate or generalized coordinate or modal coordinate. The $\mathbf{u}_k$ is the eigenvector or **modal vector**, and n is the number of modes corresponding to the number of freedom of the vibratory system. Note that, from the method of separation, since $\ddot{f}(t) + \omega^2 f(t) = 0$, hence the general solution is $f(t) = C \cos(\omega t - \phi)$, where ω is natural frequency. Thus,

$$\mathbf{x} = C_1 \cos(\omega_1 t - \phi_1) \mathbf{u}_1 + C_2 \cos(\omega_2 t - \phi_2) \mathbf{u}_2 \quad \textbf{(modal expansion)}$$

The constants C_1, C_2, ϕ_1, ϕ_2 are determined from the initial conditions.

The free response of conservative systems is a superposition of the natural modes (normal modes) multiplied by the natural coordinate (generalized coordinate). This is the basics of the **modal analysis** approach and forms a basis of the experimental modal analysis (EMA). Note that in the space of the modal coordinates, the system of equation is **decoupled**

$$\ddot{q}_k(t) + \omega_k^2 q_k(t) = 0$$

$$q_k(t) = C_k \cos(\omega_k t - \phi_k)$$

Consider $\mathbf{M}\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = 0$. Applying a transformation of the coordinate of the form

$$\mathbf{x} = \mathbf{U}\mathbf{q}$$

where, $\mathbf{U} = [\mathbf{u}_1 \ \mathbf{u}_2]$ is the modal matrix

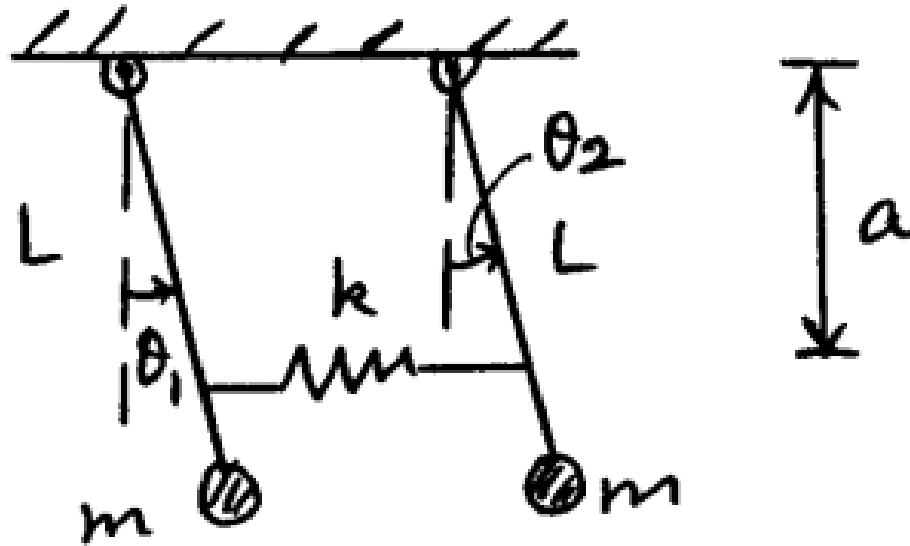
$$\mathbf{U}^T \mathbf{M} \mathbf{U} \ddot{\mathbf{q}} + \mathbf{U}^T \mathbf{K} \mathbf{U} \mathbf{q} = 0$$

where, $\mathbf{U}^T \mathbf{M} \mathbf{U} = \mathbf{I}$ is the mode normalization condition, resulting in

$$\ddot{\mathbf{q}} + \begin{bmatrix} \omega_1^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \omega_n^2 \end{bmatrix} \mathbf{q} = 0 \quad \text{Decoupling technique, leading to a set of decoupled SDOF systems}$$

2. Free Vibration Exhibiting Beating Phenomenon

When two identical SDOF systems are connected by a weak spring, then $\omega_1 \approx \omega_2$. This is called weakly coupled systems phenomena such as mode localization can occur. There is a lot of research literature on mode localization. The response to initial conditions exhibits a beating phenomenon, whereas the displacement of one pendulum is maximum while the other is minimum. Consider the following vibratory system where the spring is attached at a distance a from the pivot on each end,



The governing equation is shown as follow,

$$\begin{bmatrix} mL^2 & 0 \\ 0 & mL^2 \end{bmatrix} \ddot{\boldsymbol{\theta}} + \begin{bmatrix} mgL + ka^2 & -ka^2 \\ -ka^2 & mgL + ka^2 \end{bmatrix} \boldsymbol{\theta}(t) = 0$$

where $\boldsymbol{\theta} = \{\theta_1 \ \theta_2\}^T$, when $k \rightarrow 0$, coupling disappears and we have two independent SDOF oscillators.

The natural frequencies are: $\omega_1 = \sqrt{\frac{g}{L}}$, $\omega_2 = \sqrt{\frac{g}{L} + \frac{2ka^2}{mL^2}}$.

The normal modes (not normalized yet) are obtained as:

$\mathbf{u}_1 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$, **synchronous mode**, $\mathbf{u}_2 = \begin{Bmatrix} 1 \\ -1 \end{Bmatrix}$ **asynchronous mode** **(understand the physics)**

$$\boldsymbol{\theta}(t) = \sum_{k=1}^2 q_k(t) \mathbf{u}_k$$

Let $\theta_1(0) = \theta_0$, $\theta_2(0) = 0$, $\dot{\theta}_1(0) = 0$, $\dot{\theta}_2(0) = 0$. Then we obtain

$$\theta_1(t) = \theta_0 \cos\left(\frac{\omega_2 - \omega_1}{2} t\right) \cos\left(\frac{\omega_2 + \omega_1}{2} t\right)$$

$$\theta_2(t) = \theta_0 \sin\left(\frac{\omega_2 - \omega_1}{2} t\right) \sin\left(\frac{\omega_2 + \omega_1}{2} t\right)$$

The following pages provide the detailed derivations for the weakly coupled pendulum problem:

Lagrange's equations of motion

$$T = \frac{1}{2} m (\ell \dot{\theta}_1)^2 + \frac{1}{2} m (\ell \dot{\theta}_2)^2$$

$$V = -mg\ell \cos \theta_1 - mg\ell \cos \theta_2 + \underbrace{\frac{1}{2} k (a \sin \theta_1 - a \sin \theta_2)^2}_{\frac{ka^2}{2} (\sin \theta_1 - \sin \theta_2)^2}$$

$\sin \theta_1 \approx \theta_1$
 $\sin \theta_2 \approx \theta_2$

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_k} \right) - \frac{\partial T}{\partial q_k} + \frac{\partial V}{\partial q_k} = 0, \quad k=1, 2$$

$$\theta_1: \quad \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\theta}_1} \right) - \frac{\partial T}{\partial \theta_1} + \frac{\partial V}{\partial \theta_1} = 0$$

$$\Rightarrow m\ell^2 \ddot{\theta}_1 - 0 + mg\ell \sin \theta_1 + \frac{ka^2}{2} (2) (\sin \theta_1 - \sin \theta_2) (\cos \theta_1) = 0$$

$$\theta_2: \quad \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\theta}_2} \right) - \frac{\partial T}{\partial \theta_2} + \frac{\partial V}{\partial \theta_2} = 0$$

$$\Rightarrow m\ell^2 \ddot{\theta}_2 - 0 + mg\ell \sin \theta_2 + \frac{ka^2}{2} (2) (\sin \theta_1 - \sin \theta_2) (-\cos \theta_2) = 0$$

Linearised equations of motion

For small angles (linear vibration),

$$ml^2 \ddot{\theta}_1 + mgl \theta_1 + ka^2 (\theta_1 - \theta_2) = 0$$

$$ml^2 \ddot{\theta}_2 + mgl \theta_2 + ka^2 (\theta_2 - \theta_1) = 0$$

$$\underbrace{\begin{bmatrix} ml^2 & 0 \\ 0 & ml^2 \end{bmatrix}}_M \underbrace{\begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix}}_{\ddot{\mathbf{x}}} + \underbrace{\begin{bmatrix} mgl+ka^2 & -ka^2 \\ -ka^2 & mgl+ka^2 \end{bmatrix}}_K \underbrace{\begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix}}_{\mathbf{x}} = 0$$

mass matrix

Stiffness matrix

Formulation of the eigenvalue problem

Eigenvalue problem $\underline{K} \underline{u} = \lambda \underline{M} \underline{u}$

$$\Rightarrow (\underline{K} - \lambda \underline{M}) \underline{u} = 0$$

Require that $\det(\underline{K} - \lambda \underline{M}) = 0$ frequency eqn

Solve for λ from the frequency eqn, where $\lambda = \omega^2$

In MATLAB, use the function "eig"

$$\det \left(\begin{bmatrix} mgl+ka^2 & -ka^2 \\ -ka^2 & mgl+ka^2 \end{bmatrix} - \lambda \begin{bmatrix} ml^2 & 0 \\ 0 & ml^2 \end{bmatrix} \right) = 0$$

$$\det \left(\begin{bmatrix} \frac{g}{l} + \frac{ka^2}{ml^2} & -\frac{ka^2}{ml^2} \\ -\frac{ka^2}{ml^2} & \frac{g}{l} + \frac{ka^2}{ml^2} \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) = 0$$

$$\text{Let } \alpha = \frac{g}{l}, \quad \beta = \frac{ka^2}{ml^2}$$

$$\det \left(\begin{bmatrix} \alpha + \beta - \lambda & -\beta \\ -\beta & \alpha + \beta - \lambda \end{bmatrix} \right) = 0$$

$$\Rightarrow (\alpha + \beta - \lambda)^2 - \beta^2 = 0 \Rightarrow \alpha + \beta - \lambda = \pm \beta$$

$$\begin{aligned} \Rightarrow \lambda_1 &= \alpha + \beta - \beta = \alpha \\ \lambda_2 &= \alpha + \beta + \beta = \alpha + 2\beta \end{aligned} \quad \left. \vphantom{\begin{aligned} \Rightarrow \lambda_1 &= \alpha + \beta - \beta = \alpha \\ \lambda_2 &= \alpha + \beta + \beta = \alpha + 2\beta \end{aligned}} \right\} \lambda_2 > \lambda_1$$

$$\omega_1 = \sqrt{\lambda_1} = \sqrt{\frac{g}{l}}, \quad \omega_2 = \sqrt{\lambda_2} = \sqrt{\frac{g}{l} + \frac{2ka^2}{ml^2}}$$

$$K \underline{\tilde{u}}_1 = \lambda_1 M \underline{\tilde{u}}_1 \Rightarrow (K - \lambda_1 M) \underline{\tilde{u}}_1 = 0$$

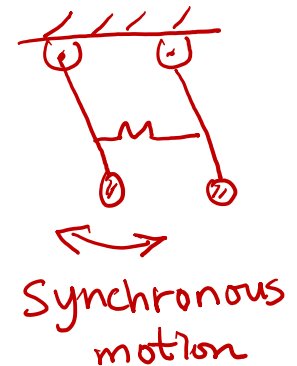
$$\Rightarrow \begin{bmatrix} mgl + ka^2 - mgl & -ka^2 \\ -ka^2 & mgl + ka^2 - mgl \end{bmatrix} \underline{\tilde{u}}_1 = 0 \Rightarrow \underline{\tilde{u}}_1 = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$$

symmetric mode
(synchronous)

Normalization of modes $\underline{\tilde{u}}_1^T M \underline{\tilde{u}}_1 = 1$, let $\underline{u}_1 = c_1 \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$

$$c_1 \begin{Bmatrix} 1 & 1 \end{Bmatrix} \begin{bmatrix} ml^2 & 0 \\ 0 & ml^2 \end{bmatrix} c_1 \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} = 1$$

$$\Rightarrow c_1^2 (2ml^2) = 1 \Rightarrow c_1 = \frac{1}{\sqrt{2ml^2}} \Rightarrow \underline{\tilde{u}}_1 = \frac{1}{\sqrt{2ml^2}} \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} \text{ first normal mode}$$



$$K \underline{\tilde{u}}_2 = \lambda_2 M \underline{\tilde{u}}_2 \Rightarrow (K - \lambda_2 M) \underline{\tilde{u}}_2 = 0$$

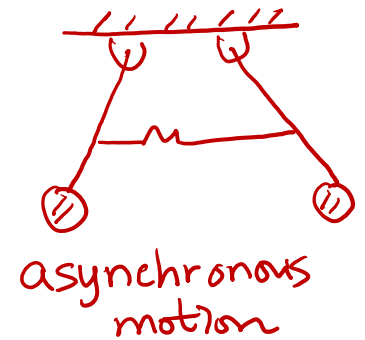
$$\begin{bmatrix} mgl + ka^2 & -mgl - 2ka^2 & -ka^2 \\ -ka^2 & mgl + ka^2 - mgl - 2ka^2 \end{bmatrix} \underline{\tilde{u}}_2 = 0 \Rightarrow \underline{\tilde{u}}_2 = \begin{Bmatrix} 1 \\ -1 \end{Bmatrix}$$

asymmetric mode
(asynchronous)

Normalization of modes $\underline{\tilde{u}}_2^T M \underline{\tilde{u}}_2 = 1$, let $\underline{\tilde{u}}_2 = c_2 \begin{Bmatrix} 1 \\ -1 \end{Bmatrix}$

$$c_2 \begin{Bmatrix} 1 & -1 \end{Bmatrix} \begin{bmatrix} ml^2 & 0 \\ 0 & ml^2 \end{bmatrix} c_2 \begin{Bmatrix} 1 \\ -1 \end{Bmatrix} = 1$$

$$\Rightarrow c_2^2 (2ml^2) = 1 \Rightarrow c_2 = \frac{1}{\sqrt{2ml^2}} \Rightarrow \underline{\tilde{u}}_2 = \frac{1}{\sqrt{2ml^2}} \begin{Bmatrix} 1 \\ -1 \end{Bmatrix} \quad \begin{array}{l} \text{second} \\ \text{normal} \\ \text{mode} \end{array}$$



The free response solution can be easily obtained by decoupling the differential eqns by the normal mode expansion method.

Consider the eqn

$$M \ddot{\underline{x}} + K \underline{x} = 0 \quad \text{where } \underline{x} = \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} \text{ for this problem}$$

Let the solution be

$$\underline{x} = \sum_{k=1}^2 q_k(t) \underline{u}_k$$

modal expansion and separation of variables concept

or write $\underline{x} = U \underline{q}$, where $\underline{q} = \begin{Bmatrix} q_1 \\ q_2 \end{Bmatrix}$, $U = [\underline{u}_1 \ \underline{u}_2]$

$$\Rightarrow U^T (M U \ddot{\underline{q}} + K U \underline{q}) = 0 \Rightarrow \ddot{\underline{q}} + \begin{bmatrix} \omega_1^2 & 0 \\ 0 & \omega_2^2 \end{bmatrix} \underline{q} = 0$$

$$\Rightarrow q_k(t) = A_k \cos \omega_k t + B_k \sin \omega_k t, \quad k=1,2 \quad (\text{undamped system})$$

The initial conditions for the generalized coordinates are

$$\begin{aligned}\underline{\tilde{x}}_0 = \underline{\tilde{x}}(0) &\Rightarrow \underline{\tilde{x}}_0 = U \underline{\tilde{q}}(0) \Rightarrow U^T M \underline{\tilde{x}}_0 = U^T M U \underline{\tilde{q}}(0) \\ &\Rightarrow \underline{\tilde{q}}_0 = \underline{\tilde{q}}(0) = U^T M \underline{\tilde{x}}_0\end{aligned}$$

Similarly, $\dot{\underline{\tilde{q}}}(0) = U^T M \underline{\tilde{v}}_0$, where $\underline{\tilde{v}}_0 = \dot{\underline{\tilde{x}}}(0)$

The general solution for $\underline{\tilde{q}}_k(t)$ is

$$q_1(t) = q_{10} \cos \omega_1 t + \frac{\dot{q}_{10}}{\omega_1} \sin \omega_1 t, \quad q_2(t) = q_{20} \cos \omega_2 t + \frac{\dot{q}_{20}}{\omega_2} \sin \omega_2 t$$

Define $\frac{\omega_B}{2} = \frac{\omega_2 - \omega_1}{2} \approx \frac{1}{2} \frac{k}{m} \frac{a^2}{\sqrt{gL^3}} \ll \sqrt{\frac{g}{L}}$ (ω_B is the beating frequency)

$$\omega_{av} = \frac{\omega_1 + \omega_2}{2} = \sqrt{\frac{g}{L}} + \frac{1}{2} \frac{k}{m} \frac{a^2}{\sqrt{gL^3}}$$

The time between two peaks of the modulated (outside) envelope is given by $\frac{T}{2} = \frac{2\pi}{\omega_B}$; T is the period of the amplitude-modulated envelope.

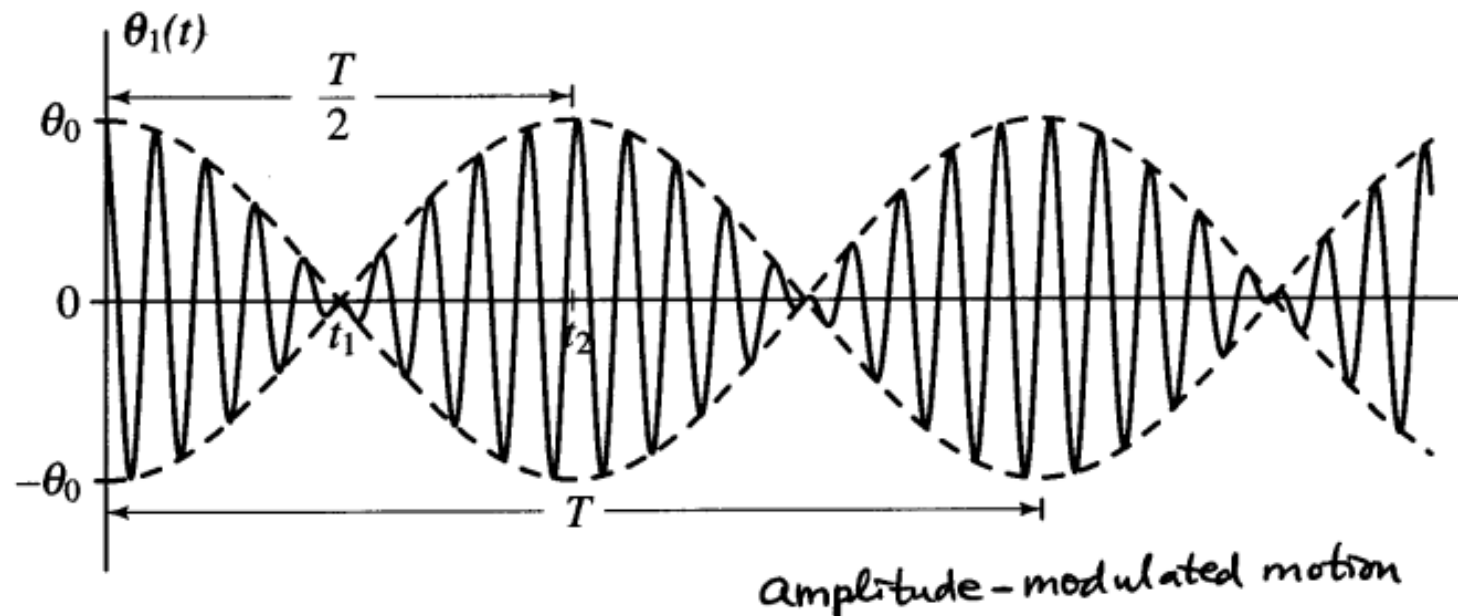
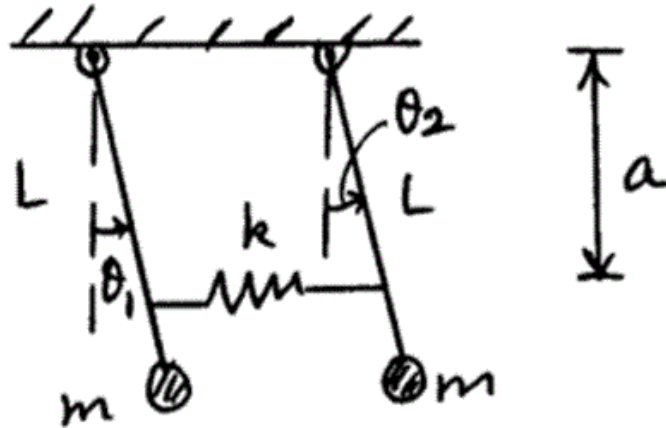


Fig.5 Response of the pendulums demonstrating the beat phenomenon

Example calculations



Consider the case of
 $a = \frac{L}{2}$, $\frac{k}{m} = \frac{2g}{L}$

We have obtained:

$$\omega_1 = \sqrt{\lambda_1} = \sqrt{\frac{g}{L}}$$

$$\omega_2 = \sqrt{\lambda_2} = \sqrt{\frac{g}{L} + \frac{2ka^2}{mL^2}}$$

$$\frac{2ka^2}{mL^2} = 2\left(\frac{2g}{L}\right)\frac{(L/2)^2}{L^2} = \frac{g}{L} \Rightarrow \omega_2 = \sqrt{\frac{2g}{L}}$$

$$\begin{bmatrix} mL^2 & 0 \\ 0 & mL^2 \end{bmatrix} \begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix} + \begin{bmatrix} mgl + ka^2 & -ka^2 \\ -ka^2 & mgl + ka^2 \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = 0$$

Substitute the parameters, $a = \frac{l}{2}$, $\frac{k}{m} = \frac{2g}{l}$

$$\Rightarrow m l^2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix} + \begin{bmatrix} mgl + \frac{2gm}{l} \left(\frac{l^2}{4}\right) & -\frac{2gm}{l} \left(\frac{l^2}{4}\right) \\ -\frac{2gm}{l} \left(\frac{l^2}{4}\right) & mgl + \frac{2gm}{l} \left(\frac{l^2}{4}\right) \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = 0$$

$$\Rightarrow m l^2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix} + \begin{bmatrix} 1.5 mgl & -0.5 mgl \\ -0.5 mgl & 1.5 mgl \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix} + \frac{g}{l} \begin{bmatrix} 1.5 & -0.5 \\ -0.5 & 1.5 \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = 0$$

$$K \underline{u} = \lambda M \underline{u} \Rightarrow \frac{g}{l} \begin{bmatrix} 1.5 & -0.5 \\ -0.5 & 1.5 \end{bmatrix} \underline{u} = \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \underline{u}$$

$$\text{define } \tilde{\lambda} = \frac{\lambda l}{g} \Rightarrow \underbrace{\begin{bmatrix} 1.5 & -0.5 \\ -0.5 & 1.5 \end{bmatrix}}_{\text{new K}} \underline{u} = \tilde{\lambda} \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{\text{new M}} \underline{u}$$

↓
non-dimensional

Formulate the vibration problem in non-dimensional form

Solve EVP by MATLAB "eig"

Type "help eig"

[use lookfor command if you
do not know name of function]

$$K = [1.5 \ -0.5; -0.5 \ 1.5];$$

$$M = \text{eye}(2);$$

$$[V, D] = \text{eig}(K, M) \rightarrow$$

modal matrix

$$V = \begin{bmatrix} -0.7071 & -0.7071 \\ -0.7071 & 0.7071 \end{bmatrix},$$

$$D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

eigenvalues

$$\hat{\lambda}_1 = 1$$

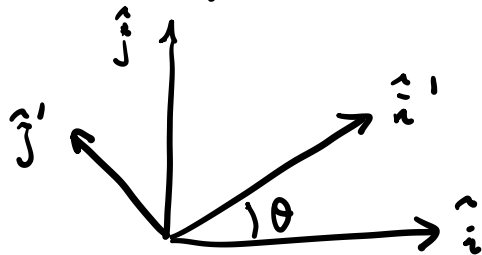
$$\hat{\lambda}_2 = 2$$

$$D = \begin{bmatrix} \hat{\lambda}_1 & & \\ & \ddots & \\ & & \hat{\lambda}_n \end{bmatrix}$$

← solve $\det(K - \hat{\lambda} M) = 0$

eigenvectors
or modal vectors

Coordinate transformation



$$\begin{Bmatrix} \hat{i}' \\ \hat{j}' \end{Bmatrix} = \underbrace{\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}}_{\det = 1} \begin{Bmatrix} \hat{i} \\ \hat{j} \end{Bmatrix}$$

The actual frequencies for this system, obtained by hand calculations, are

$$\omega_1 = \sqrt{\lambda_1} = \sqrt{\frac{g}{l}}, \quad \omega_2 = \sqrt{\lambda_2} = \sqrt{\frac{g}{l} + \frac{2ka^2}{ml^2}}$$

Since $a = \frac{l}{2}$, $\frac{k}{m} = 2\frac{g}{l}$, $\omega_2 = \sqrt{\frac{g}{l} + 2\left(\frac{2g}{l}\right)\left(\frac{l^2}{4}\right)} = \sqrt{2\frac{g}{l}}$

Same

MATLAB returns:

$$\tilde{\lambda}_1 = 1 \Rightarrow \lambda_1 = \frac{g}{l} \quad \tilde{\lambda}_1 = \frac{g}{l} \Rightarrow \omega_1 = \sqrt{\frac{g}{l}}$$

$$\tilde{\lambda}_2 = 2 \Rightarrow \lambda_2 = \frac{g}{l} \quad \tilde{\lambda}_2 = \frac{2g}{l} \Rightarrow \omega_2 = \sqrt{\frac{2g}{l}}$$

Go back to the original problem:

$$ml^2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix} + \begin{bmatrix} 1.5 mgl & -0.5 mgl \\ -0.5 mgl & 1.5 mgl \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = 0$$

Define normal modes

$$\underline{u}_1^T M \underline{u}_1 = 1, \text{ let } \underline{u}_1 = \alpha_1 \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} \Rightarrow \underline{u}_1 = \frac{1}{\sqrt{2ml^2}} \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$$

$$\underline{u}_2^T M \underline{u}_2 = 1, \text{ let } \underline{u}_2 = \alpha_2 \begin{Bmatrix} -1 \\ 1 \end{Bmatrix} \Rightarrow \underline{u}_2 = \frac{1}{\sqrt{2ml^2}} \begin{Bmatrix} -1 \\ 1 \end{Bmatrix}$$

$$\text{Modal matrix } U = [\underline{u}_1 \ \underline{u}_2] = \frac{1}{\sqrt{2ml^2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

Given initial conditions, $\underline{x}(0) = \underline{x}_0$, $\dot{\underline{x}}_0 = \underline{v}_0$

$$\underline{x} = U \underline{q} \Rightarrow \underline{x}_0 = U \underline{q}_0 \Rightarrow \underline{q}_0 = U^{-1} \underline{x}_0$$

$$U^T M U = I \Rightarrow U^T M = U^{-1} \Rightarrow \underline{q}_0 = U^T M \underline{x}_0$$

$$\text{Also } \dot{\underline{q}}_0 = U^T M \underline{v}_0$$

$$\text{For example: } \underline{x}(0) = \begin{Bmatrix} \theta_0 \\ 2\theta_0 \end{Bmatrix}, \underline{v}_0 = 0$$

$$\Rightarrow \underline{q}_0 = \frac{1}{\sqrt{2ml^2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} ml^2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{Bmatrix} \theta_0 \\ 2\theta_0 \end{Bmatrix} = \sqrt{\frac{ml^2}{2}} \begin{Bmatrix} 3\theta_0 \\ \theta_0 \end{Bmatrix}$$

$$\text{Suppose } \underline{x}_0 = \begin{Bmatrix} \theta_0 \\ \theta_0 \end{Bmatrix}$$

$$\Rightarrow \underline{q}_0 = \sqrt{\frac{ml^2}{2}} \begin{Bmatrix} 2\theta_0 \\ 0 \end{Bmatrix} \leftarrow \text{the initial condition for the second mode is zero!}$$

Finally, obtain the response solution:

$$\ddot{q}_k(t) + \omega_k^2 q_k(t) = 0$$

$$\Rightarrow q_1(t) = \sqrt{\frac{ml^2}{2}} (3\theta_0) \cos \omega_1 t, \quad q_2(t) = \sqrt{\frac{ml^2}{2}} \theta_0 \cos \omega_2 t$$

$$\begin{aligned} \tilde{x} &= q_1 \tilde{u}_1 + q_2 \tilde{u}_2 = \sqrt{\frac{ml^2}{2}} (3\theta_0) \cos \omega_1 t \left(\frac{1}{\sqrt{2ml^2}} \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} \right) \\ &\quad + \sqrt{\frac{ml^2}{2}} (\theta_0) \cos \omega_2 t \left(\frac{1}{\sqrt{2ml^2}} \begin{Bmatrix} -1 \\ 1 \end{Bmatrix} \right) \\ &= \frac{3\theta_0}{2} \cos \omega_1 t \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} + \frac{\theta_0}{2} \cos \omega_2 t \begin{Bmatrix} -1 \\ 1 \end{Bmatrix} \end{aligned}$$

3. Response of 2DOF systems to harmonic excitations

Consider $\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{F}(t)$

Let $\mathbf{F}(t) = \bar{\mathbf{F}}e^{i\omega t}$ (complex notation)
 $\mathbf{x}(t) = \bar{\mathbf{X}}(\omega)e^{i\omega t}$ ($\bar{\mathbf{X}}(\omega)$ is a complex-valued vector)

By direct substitution, we get

$$\bar{\mathbf{X}}(\omega) = \mathbf{Z}^{-1}(i\omega) \bar{\mathbf{F}}, \quad \mathbf{Z}(i\omega) = -\omega^2 \mathbf{M} + i\omega \mathbf{C} + \mathbf{K}$$

$\mathbf{Z}(i\omega)$ is analogous to the impedance function for SDOF systems.

$$\bar{X}_1(\omega) = \frac{Z_{22}(i\omega)F_1 - Z_{12}(i\omega)F_2}{Z_{11}(i\omega)Z_{22}(i\omega) - Z_{12}^2(i\omega)}, \quad \bar{X}_2(\omega) = \frac{Z_{11}(i\omega)F_2 - Z_{12}(i\omega)F_1}{Z_{11}(i\omega)Z_{22}(i\omega) - Z_{12}^2(i\omega)}$$

For undamped system, $\mathbf{Z}(i\omega) = -\omega^2 \mathbf{M} + \mathbf{K}$ is a real matrix function called the impedance matrix. This approach is similar to the transfer function method (frequency response function FRF) but is not practical to implement for two or higher DOF systems due to the inverse. The proper approach is the modal analysis method.

Example 1.4

Consider such a system is shown in figure6, the mass $m_1 = m, m_2 = 2m; L_1 = L_2 = L, L_3 = 0.5L$.

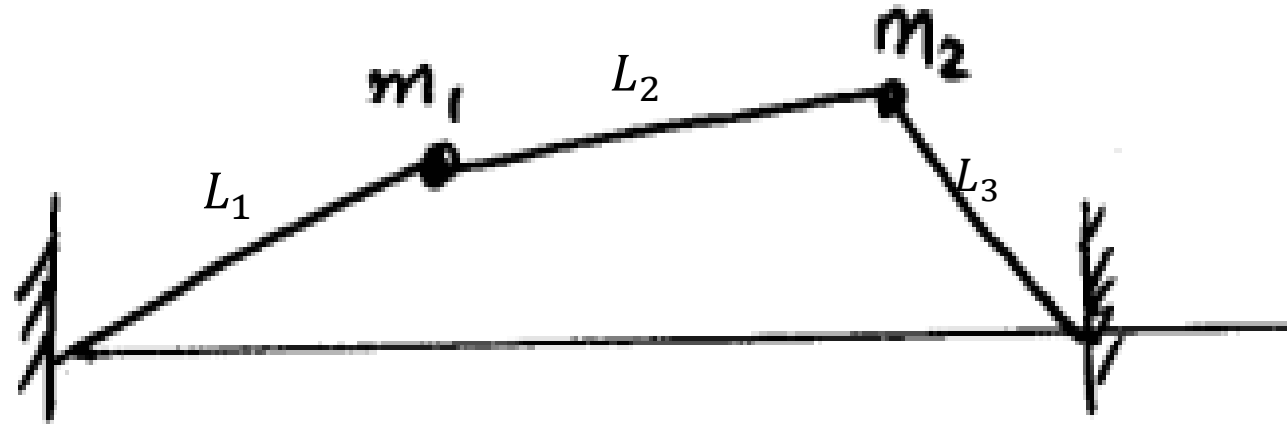


Fig. 6 Schematic diagram

Solution:

We can easily get

$$k_{11} = \frac{2T}{L}, k_{22} = \frac{3T}{L}, k_{12} = \frac{-T}{L}$$

Let $F_1 \neq 0, F_2 \neq 0$. The response can be shown to be

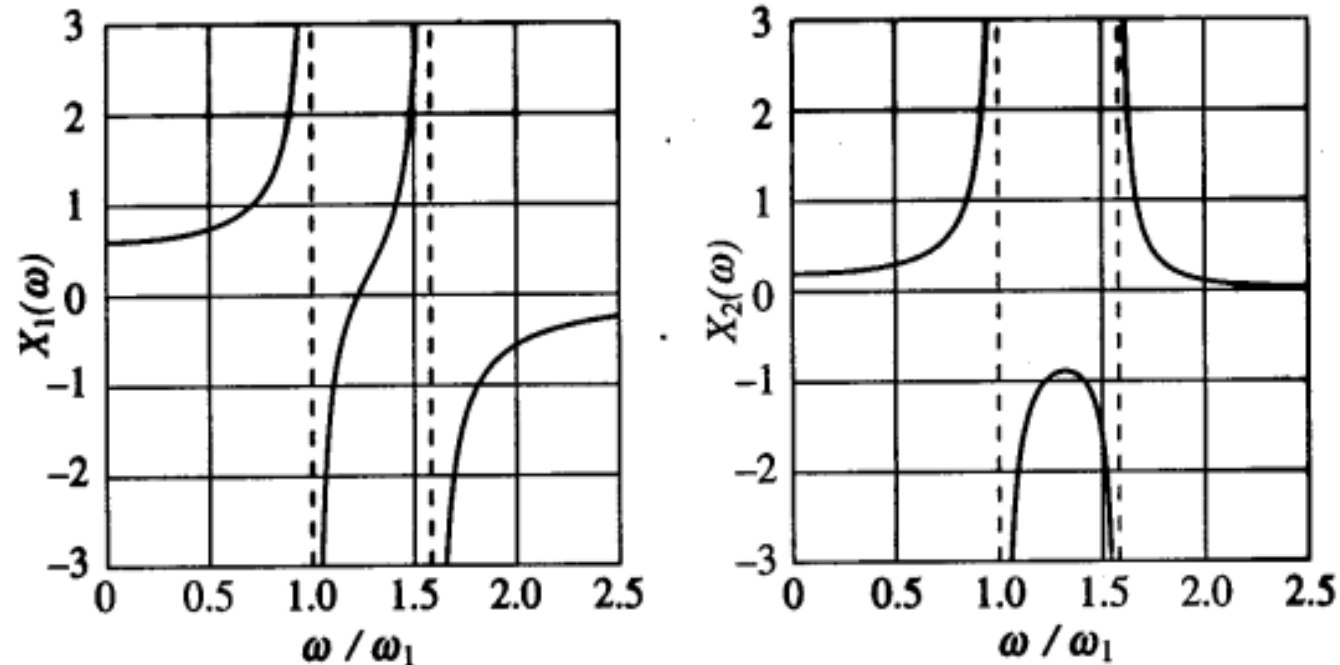
$$\bar{X}_1(\omega) = \frac{(3k - 2m\omega^2)F_1}{2m^2\omega^4 - 7mk\omega^2 + 5k^2}, \bar{X}_2(\omega) = \frac{kF_1}{2m^2\omega^4 - 7mk\omega^2 + 5k^2}$$

where $k = \frac{T}{L}$.

The denominator is $\Delta(\omega^2) = 2m(\omega^2 - \omega_1^2)(\omega^2 - \omega_2^2)$

Characteristic equation

Frequency response curves for an undamped 2DOF system is shown below.



Example: Vibration absorber

$$\begin{aligned} F_1 &\neq 0 \\ F_2 &= 0 \end{aligned}$$

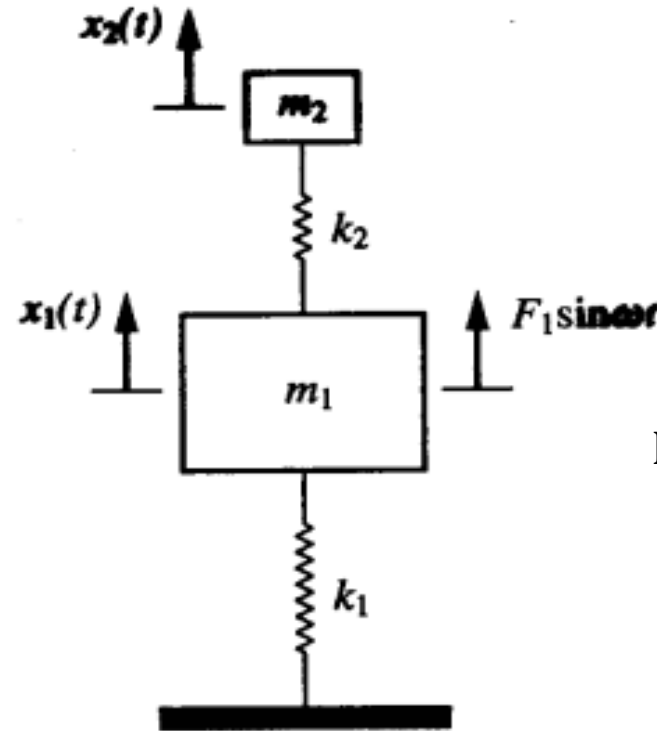


Fig. 7 vibration absorber

Solution:

main system: m_1 and k_1 .

Absorber: m_2 and k_2 .

The governing equations are

$$\begin{aligned} m_1 \ddot{x}_1 + (k_1 + k_2)x_1 - k_2 x_2 &= F_1 \sin \omega t \\ m_2 \ddot{x}_2 - k_2 x_1 + k_2 x_2 &= 0 \end{aligned}$$

For undamped system: $x_1(t) = \bar{X}_1 \sin \omega t$, $x_2(t) = \bar{X}_2 \sin \omega t$ (forced response solutions)

$$\begin{cases} \frac{(k_2 - \omega^2 m_2)F_1}{(k_1 + k_2 - \omega^2 m_1)(k_2 - \omega^2 m_2) - k_2^2} \\ \frac{k_2 F_1}{(k_1 + k_2 - \omega^2 m_1)(k_2 - \omega^2 m_2) - k_2^2} \end{cases}$$

Let $\omega_n = \sqrt{\frac{k_1}{m_1}}$, $\omega_a = \sqrt{\frac{k_2}{m_2}}$, $x_{st} = \frac{F_1}{k_1}$, $\mu = \frac{m_2}{m_1}$.

Then
$$\begin{cases} \bar{X}_1 = \frac{[1 - (\frac{\omega}{\omega_a})^2] x_{st}}{[1 + \mu(\omega_a/\omega_n)^2 - (\omega/\omega_n)^2][1 - (\frac{\omega}{\omega_a})^2] - \mu(\frac{\omega_a}{\omega_n})^2} \\ \bar{X}_2 = \frac{x_{st}}{[1 + \mu(\omega_a/\omega_n)^2 - (\omega/\omega_n)^2][1 - (\frac{\omega}{\omega_a})^2] - \mu(\frac{\omega_a}{\omega_n})^2} \end{cases}$$

At $\omega = \omega_a$, $\bar{X}_1 = 0$ (vibration absorption if absorber frequency $\approx \omega$)

$$\text{At } \omega = \omega_n, \bar{X}_2 = -\left(\frac{\omega_n}{\omega_a}\right)^2 \frac{x_{st}}{\mu} = \frac{-F_1}{k_2} \quad x_2(t) = -F_1 \sin \omega t$$

$$\text{force in absorber spring } k_2 x_2(t) = -F_1 \sin \omega t$$

For response to periodic excitation, use superposition method as before.

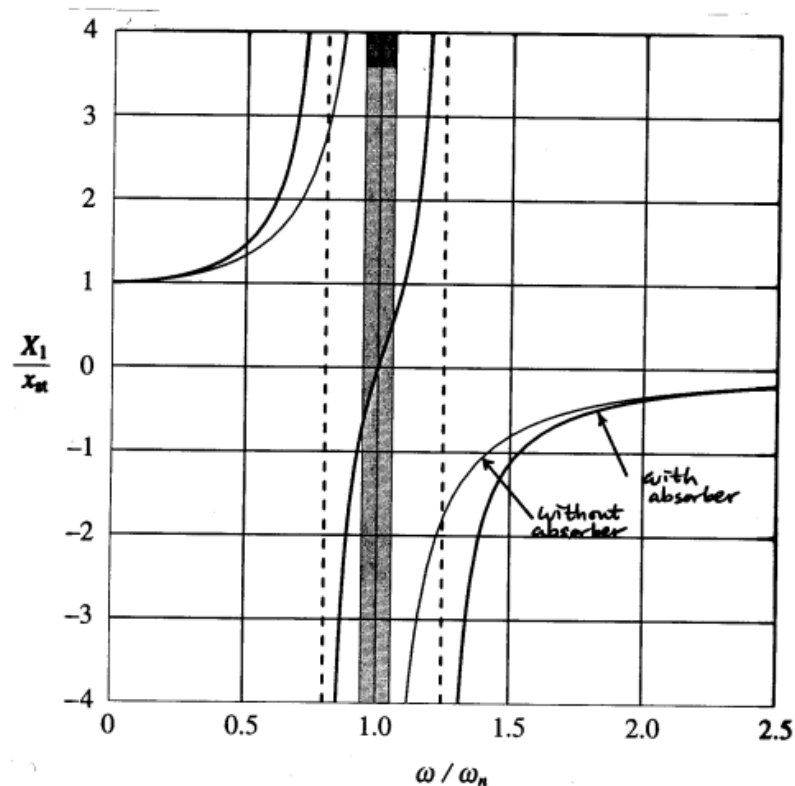


Fig. 8 Frequency response curve for the main mass

4. Response of 2DOF systems to Nonperiodic Excitations

Consider undamped systems for which the eigenvectors and eigenvalues are real and can be obtained easily. Damped systems will be considered in the content of MDOF systems.

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{F}(t)$$

Let $\mathbf{x} = \eta_1(t)\mathbf{u}_1 + \eta_2(t)\mathbf{u}_2$

$$\mathbf{U} = [\mathbf{u}_1 \ \mathbf{u}_2] \rightarrow U^T \mathbf{M} \mathbf{U} = \begin{bmatrix} m'_{11} & 0 \\ 0 & m'_{22} \end{bmatrix}, U^T \mathbf{K} \mathbf{U} = \begin{bmatrix} m'_{11}\omega_1^2 & 0 \\ 0 & m'_{22}\omega_2^2 \end{bmatrix}$$

$$\rightarrow \begin{cases} m'_{11}\ddot{\eta}_1(t) + m'_{11}\eta_1(t) = N_1(t) = u_1^T F(t) \\ m'_{22}\ddot{\eta}_2(t) + m'_{22}\eta_2(t) = N_2(t) = u_2^T F(t) \end{cases}$$

$$\rightarrow \begin{cases} \eta_1(t) = \frac{1}{m'_{11}\omega_1} \int_0^t N_1(t-\tau) \sin \omega_1 \tau \mu(\tau) d\tau = \frac{1}{m'_{11}\omega_1} \int_0^t N_1(t-\tau) \sin \omega_1 \tau d\tau \\ \eta_2(t) = \frac{1}{m'_{22}\omega_2} \int_0^t N_2(t-\tau) \sin \omega_2 \tau \mu(\tau) d\tau = \frac{1}{m'_{22}\omega_2} \int_0^t N_2(t-\tau) \sin \omega_2 \tau d\tau \end{cases}$$

For simplicity, normalise the eigenvectors using the mass-normalization approach:

$$\mathbf{u}_i^T \mathbf{M} \mathbf{u}_j = \delta_{ij}; \mathbf{U}^T \mathbf{M} \mathbf{U} = \mathbf{I}, \mathbf{U}^T \mathbf{K} \mathbf{U} = \begin{bmatrix} \omega_1^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \omega_n^2 \end{bmatrix}$$

When the nature of the force vector $F(t)$ preclude the use of analytical evaluation such as the convolution method (integration), we can proceed with convolution sums (numerical sums). Denote

$$\mathbf{x}(n) = \eta_1(n)\mathbf{u}_1 + \eta_2(n)\mathbf{u}_2, \quad n = 1, 2, \dots$$

where $\eta_j(n)$ are the modal coordinates at $t = nT_s$; T_s is the sampling period. From pervious results, we have

$$\eta_1(n) = \sum_{k=0}^n N_1(k)g_1(n-k), \quad \eta_2(n) = \sum_{k=0}^n N_2(k)g_2(n-k)$$

$$g_j(n) = \frac{T_s}{m'_{jj}\omega_j^2} \sin(n\omega_j T_s); j = 1, 2$$

$$\mathbf{x}(n) = T_s \sum_{k=0}^n \left\{ \left[\frac{\mathbf{u}_1^T}{m'_{11}\omega_1} F(k) \sin(n-k)\omega_1 T_s \right] \mathbf{u}_1 + \left[\frac{\mathbf{u}_2^T}{m'_{22}\omega_2} F(k) \sin(n-k)\omega_2 T_s \right] \mathbf{u}_2 \right\}$$

The normalized mass quantities are usually unity, $m'_{11} = 1$ and $m'_{22} = 1$.

Example: String-rod 2DOF system

Derive the governing equation of motion of a thin rod attached to a massless, inextensible string.

1. By Lagrange's equation

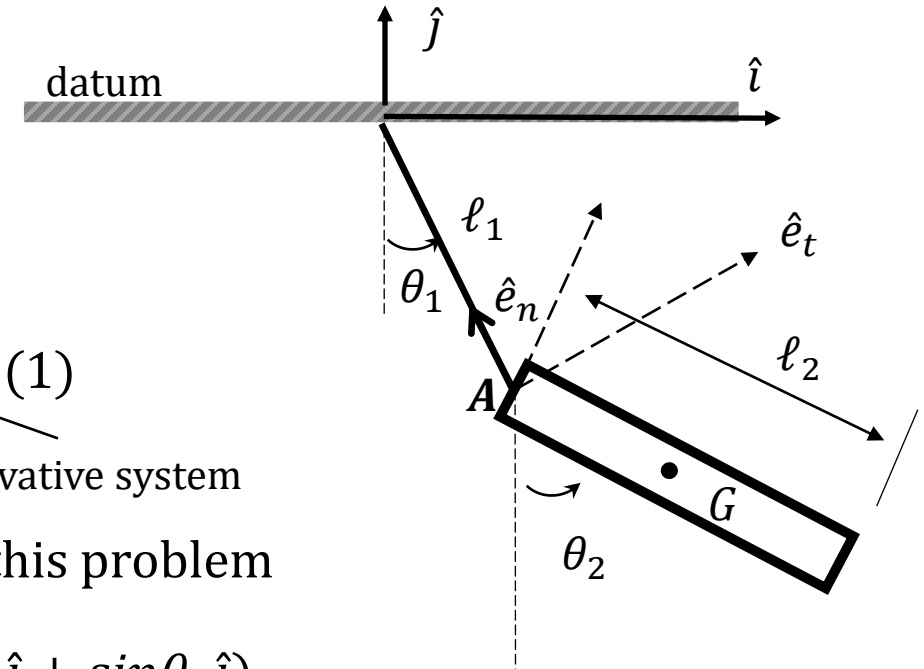
The generalized coordinates are: $q_1 = \theta_1$, $q_2 = \theta_2$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right) - \frac{\partial L}{\partial q_k} = Q_k, \quad L = T - V, \quad Q_k = 0 \quad (1)$$

Conservative system

It's probably easier to use the $\hat{i} - \hat{j}$ coordinate system for this problem

$$\begin{aligned} \vec{v}_G &= \vec{V}_A + \vec{\omega} \times \vec{r}_{G/A}, \quad \vec{v}_A = \ell_1 \dot{\theta}_1 \hat{e}_t = \ell_1 \dot{\theta}_1 (\cos\theta_1 \hat{i} + \sin\theta_1 \hat{j}) \\ \Rightarrow \vec{v}_G &= \ell_1 \dot{\theta}_1 (\cos\theta_1 \hat{i} + \sin\theta_1 \hat{j}) + \dot{\theta}_2 \hat{k} \times \frac{\ell_2}{2} (\sin\theta_2 \hat{i} - \cos\theta_2 \hat{j}) \\ \Rightarrow \vec{v}_G &= (\ell_1 \dot{\theta}_1 \cos\theta_1 + \frac{\ell_2}{2} \dot{\theta}_2 \cos\theta_2) \hat{i} + (\ell_1 \dot{\theta}_1 \sin\theta_1 + \frac{\ell_2}{2} \dot{\theta}_2 \sin\theta_2) \hat{j} \end{aligned} \quad (2)$$



Kinetic energy:

$$T = \frac{1}{2}mv_G^2 + \frac{1}{2}I_G\dot{\theta}_2^2 \quad \text{where } I_G = \frac{m\ell_2^2}{12}$$
$$\Rightarrow T = \frac{1}{2}m \left(\ell_1^2\dot{\theta}_1^2 + \frac{\ell_2^2}{4}\dot{\theta}_2^2 + \ell_1\ell_2\dot{\theta}_1\dot{\theta}_2\cos\theta_1\cos\theta_2 + \ell_1\ell_2\dot{\theta}_1\dot{\theta}_2\sin\theta_1\theta_2 \right)$$
$$\Rightarrow T = \frac{1}{2}m \left[\ell_1^2\dot{\theta}_1^2 + \frac{\ell_2^2}{4}\dot{\theta}_2^2 + \ell_1\ell_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_2 - \theta_1) \right] + \frac{m\ell_2^2}{12}\dot{\theta}_2^2 \quad (3)$$

Potential energy:

$$V = -mg \left(\ell_1\cos\theta_1 + \frac{\ell_2}{2}\cos\theta_2 \right) \quad (4)$$

θ_1 -equation:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_1} \right) = \frac{d}{dt} \left[m\ell_1^2 \dot{\theta}_1 + \frac{m}{2} \ell_1 \ell_2 \dot{\theta}_2 \cos(\theta_2 - \theta_1) \right]$$
$$\frac{\partial L}{\partial \theta_1} = \frac{m\ell_1 \ell_2}{2} \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_2 - \theta_1) - mg\ell_1 \sin \theta_1$$

Applying the Lagrange's equations from Eq. (1) gives

$$m\ell_1^2 \ddot{\theta}_1 + \frac{m\ell_1 \ell_2}{2} \ddot{\theta}_2 \cos(\theta_2 - \theta_1) - \frac{m\ell_1 \ell_2}{2} \dot{\theta}_2 \sin(\theta_2 - \theta_1) (\dot{\theta}_2 - \dot{\theta}_1) - \frac{m\ell_1 \ell_2}{2} \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_2 - \theta_1) + mg\ell_1 \sin \theta_1 = 0$$
$$\Rightarrow m\ell_1^2 \ddot{\theta}_1 + \frac{m\ell_1 \ell_2}{2} \ddot{\theta}_2 \cos(\theta_2 - \theta_1) - \frac{m\ell_1 \ell_2}{2} \dot{\theta}_2^2 \sin(\theta_2 - \theta_1) + mg\ell_1 \sin \theta_1 = 0 \quad (5)$$

For small angle of oscillation (linear vibration), we approx. $\cos \theta \approx 1$, $\sin \theta \approx \theta$, and neglect nonlinear terms. Thus, Eq. (5) become:

$$m\ell_1^2 \ddot{\theta}_1 + \frac{m\ell_1 \ell_2}{2} \ddot{\theta}_2 + mg\ell_1 \theta_1 = 0 \quad (6)$$

The coordinates are coupled by inertia

θ_2 -equation:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_2} \right) = \frac{d}{dt} \left[\frac{ml_2^2}{3} \dot{\theta}_2 + \frac{m}{2} l_1 l_2 \dot{\theta}_1 \cos(\theta_2 - \theta_1) \right]$$
$$\frac{\partial L}{\partial \theta_2} = -\frac{ml_1 l_2}{2} \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_2 - \theta_1) - \frac{mgl_2}{2} \sin \theta_2$$

Apply Lagrange's equations from Eq. (1) gives

$$\begin{aligned} \frac{ml_2^2}{3} \ddot{\theta}_2 + \frac{ml_1 l_2}{2} \ddot{\theta}_1 \cos(\theta_2 - \theta_1) - \frac{ml_1 l_2}{2} \dot{\theta}_1 \sin(\theta_2 - \theta_1) (\dot{\theta}_2 - \dot{\theta}_1) \\ + \frac{ml_1 l_2}{2} \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_2 - \theta_1) + \frac{mgl_2}{2} \sin \theta_2 = 0 \end{aligned}$$
$$\Rightarrow \frac{ml_2^2}{3} \ddot{\theta}_2 + \frac{ml_1 l_2}{2} \ddot{\theta}_1 \cos(\theta_2 - \theta_1) + \frac{ml_1 l_2}{2} \dot{\theta}_1^2 \sin(\theta_2 - \theta_1) + \frac{mgl_2}{2} \sin \theta_2 = 0 \quad (7)$$

For small angles, linearize Eq. (7) to get:

$$\frac{ml_2^2}{3} \ddot{\theta}_2 + \frac{ml_1 l_2}{2} \ddot{\theta}_1 + \frac{mgl_2}{2} \theta_2 = 0 \quad (8)$$

The coordinates are coupled by inertia

2. Application of Newton's Law :

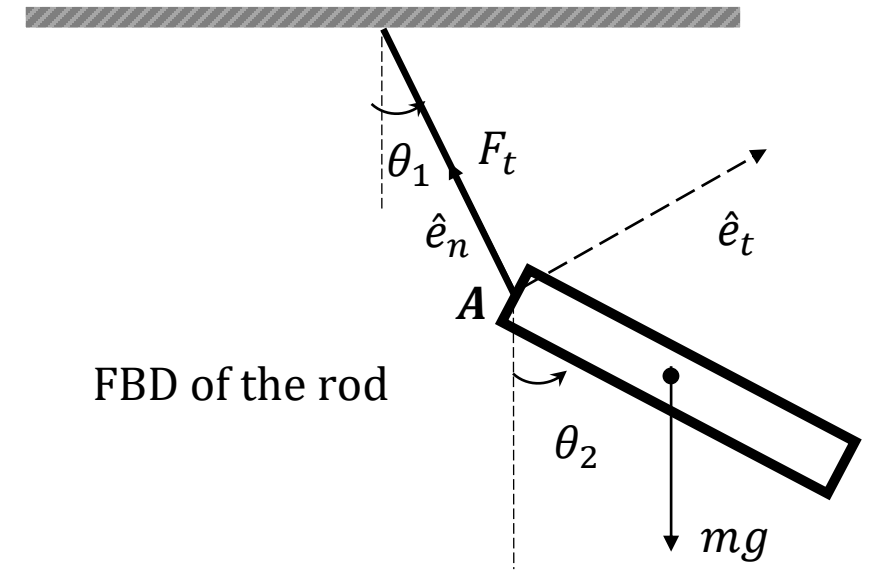
We need to generate two equation without involving the tension force F_t .

$$\begin{aligned}\vec{a}_A &= l_1 \ddot{\theta}_1 \hat{e}_t + l_1 \dot{\theta}_1^2 \hat{e}_n \\ &= l_1 \ddot{\theta}_1 (\cos\theta_1 \hat{i} + \sin\theta_1 \hat{j}) + l_1 \dot{\theta}_1^2 (-\sin\theta_1 \hat{i} + \cos\theta_1 \hat{j}) \\ &= (l_1 \ddot{\theta}_1 \cos\theta_1 - l_1 \dot{\theta}_1^2 \sin\theta_1) \hat{i} + (l_1 \ddot{\theta}_1 \sin\theta_1 + l_1 \dot{\theta}_1^2 \cos\theta_1) \hat{j}\end{aligned}$$

Note: there is probably a simple approach by applying the coordinate transformation matrix learn in dynamics

$$\vec{r}_{G/A} = \frac{l_2}{2} [\sin(\theta_2 - \theta_1) \hat{e}_t - \cos(\theta_2 - \theta_1) \hat{e}_n]$$

$$\begin{aligned}\vec{a}_G &= \vec{a}_A + \vec{\alpha} \times \vec{r}_{G/A} - \omega^2 \vec{r}_{G/A} \\ &= l_1 \ddot{\theta}_1 \hat{e}_t + l_1 \dot{\theta}_1^2 \hat{e}_n + \ddot{\theta}_2 \hat{k} \times \frac{l_2}{2} [\sin(\theta_2 - \theta_1) \hat{e}_t - \cos(\theta_2 - \theta_1) \hat{e}_n] \\ &\quad - \dot{\theta}_2^2 \frac{l_2}{2} [\sin(\theta_2 - \theta_1) \hat{e}_t - \cos(\theta_2 - \theta_1) \hat{e}_n]\end{aligned}$$



Collecting terms for $\vec{a}_G$

$$\hat{e}_t: \quad l_1 \ddot{\theta}_1 + \frac{l_2}{2} \ddot{\theta}_2 \cos(\theta_2 - \theta_1) - \frac{l_2}{2} \dot{\theta}_2^2 \sin(\theta_2 - \theta_1)$$

$$\hat{e}_n: \quad l_1 \dot{\theta}_1^2 + \frac{l_2}{2} \ddot{\theta}_2 \sin(\theta_2 - \theta_1) - \frac{l_2}{2} \dot{\theta}_2^2 \cos(\theta_2 - \theta_1)$$

$$\begin{array}{l|l} \sum F_{\text{tangential}} = m(a_G)_t & \vec{F}_g = -mg\hat{j} \\ \text{Only gravity force} & \vec{F}_g \cdot \hat{e}_t = -mg\hat{j} \cdot (\cos\theta_1\hat{i} + \sin\theta_1\hat{j}) = -mg \sin \theta_1 \\ \text{has component in this direction} & ! \end{array}$$

$$\Rightarrow -mg \sin \theta_1 = ml_1 \ddot{\theta}_1 + \frac{ml_2}{2} \ddot{\theta}_2 \cos(\theta_2 - \theta_1) - \frac{ml_2}{2} \dot{\theta}_2^2 \sin(\theta_2 - \theta_1)$$

$$\Rightarrow ml_1 \ddot{\theta}_1 + \frac{ml_2}{2} \ddot{\theta}_2 \cos(\theta_2 - \theta_1) - \frac{ml_2}{2} \dot{\theta}_2^2 \sin(\theta_2 - \theta_1) + mg \sin \theta_1 = 0 \quad \begin{array}{l} \text{This is the same as eq. (5),} \\ \text{divided by } \ell_1. \end{array}$$

For small angle of vibration, we obtain:

$$ml_1 \ddot{\theta}_1 + \frac{ml_2}{2} \ddot{\theta}_2 + mg\theta_1 = 0 \quad \text{This is the same as equation (6), divided by } l_1.$$

Consider the moment Equation apply at point A (not a fixed pt). Thus,

$$\sum M_A = I_G \alpha + (\vec{r}_{G/A} \times m \vec{a}_G) \cdot \hat{k}$$

$$\sum M_A = -mg \frac{\ell_2}{2} \sin(\theta_2)$$

$$I_G \alpha = \frac{ml_2^2}{12} \ddot{\theta}_2$$

$$\vec{r}_{G/A} \times m \vec{a}_G = \frac{ml_2}{2} [\sin(\theta_2 - \theta_1) \hat{e}_t - \cos(\theta_2 - \theta_1) \hat{e}_n] \times [(a_G)_t \hat{e}_t - (a_G)_n \hat{e}_n]$$

$$\Rightarrow (\vec{r}_{G/A} \times m \vec{a}_G) \cdot \hat{k} = \frac{ml_2}{2} [\sin(\theta_2 - \theta_1) (a_G)_n + \cos(\theta_2 - \theta_1) (a_G)_t]$$

Grouping all terms:

$$-\frac{mgl_2}{2} \sin\theta_2 = \frac{ml_2^2}{12} \ddot{\theta}_2 + \frac{ml_2}{2} [\sin(\theta_2 - \theta_1) \left(l_1 \dot{\theta}_1^2 + \frac{l_2}{2} \ddot{\theta}_2 \sin(\theta_2 - \theta_1) + \frac{l_2}{2} \dot{\theta}_2^2 \cos(\theta_2 - \theta_1) \right) + \cos(\theta_2 - \theta_1) (l_1 \ddot{\theta}_1 + \frac{l_2}{2} \ddot{\theta}_2 \cos(\theta_2 - \theta_1) - \frac{l_2}{2} \dot{\theta}_2^2 \sin(\theta_2 - \theta_1))]]$$

$$\frac{ml_2^2}{3} \ddot{\theta}_2 + \frac{ml_1 l_2}{2} \dot{\theta}_1^2 \sin(\theta_2 - \theta_1) + \frac{ml_1 l_2}{2} \ddot{\theta}_1 \cos(\theta_2 - \theta_1) + \frac{mgl_2}{2} \sin\theta_2 = 0 \quad (7)$$

This result is exactly Eq.(7)

Again, linearization leads to:

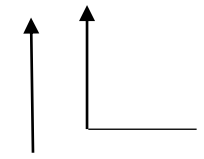
$$\frac{ml_2^2}{3} \ddot{\theta}_2 + \frac{ml_1 l_2}{2} \ddot{\theta}_1 + \frac{mgl_2}{2} \theta_2 = 0 \quad (8)$$

3. Linear Vibration Analysis (modal analysis)

From Eqs. (6) and (8), we have


$$\begin{bmatrix} ml_1^2 & \frac{ml_1 l_2}{2} \\ \frac{ml_1 l_2}{2} & \frac{ml_2^2}{3} \end{bmatrix} \begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix} + \begin{bmatrix} mgl_1 & 0 \\ 0 & \frac{mgl_2}{2} \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = 0$$
$$[M]\{\ddot{q}\} + [K]\{q\} = 0$$

This formulation guarantees symmetric $[M]$ and $[K]$. The general solution by modal analysis assumes:

$$q(t) = u e^{i\omega t}$$


Separation of variables

Temporal response

Mode shape

$$\Rightarrow -\omega^2[M]ue^{i\omega t} + [K]ue^{i\omega t} = 0$$

$$\Rightarrow \text{Eigenvalue problem: } [K]u = \lambda[M]u \text{ where } \lambda = \omega^2$$

Solve the eigenvalue problem:

$\det(K - \lambda M) = 0$ frequency equation to obtain eigenvalues

$[K]u_j = \lambda[M]u_j$ solve for the j-th eigenvector or mode shape vector

Use MATLAB to solve the eigenvalue problem (EVP) to find the frequency and mode shapes (EVP is the same as learned in Math Class).

Note: For numerical computations, we must define non-dimensional parameters in order to be able to enter numbers into MATLAB:

$$[\tilde{M}] = \begin{bmatrix} 1 & \frac{\alpha}{2} \\ \frac{\alpha}{2} & \frac{\alpha^2}{3} \end{bmatrix} \quad [\tilde{K}] = \begin{bmatrix} 1 & 0 \\ 0 & \frac{\alpha}{2} \end{bmatrix} \quad \text{where } \alpha = \frac{l_2}{l_1}$$

$$\text{Solve } [\tilde{K}]u = \tilde{\lambda}[\tilde{M}]u \quad \text{where } \tilde{\lambda} = \omega^2 \left(\frac{\ell_1}{g} \right) \quad (\text{Non-dimensionalised eigenvalue})$$

Then obtain modal response following notes.

Example: A 2DOF spring-mass system

Find the natural frequencies and mode shapes of a spring-mass system, shown in Fig. 5.6, which is constrained to move in the vertical direction only. Take $n = 1$.

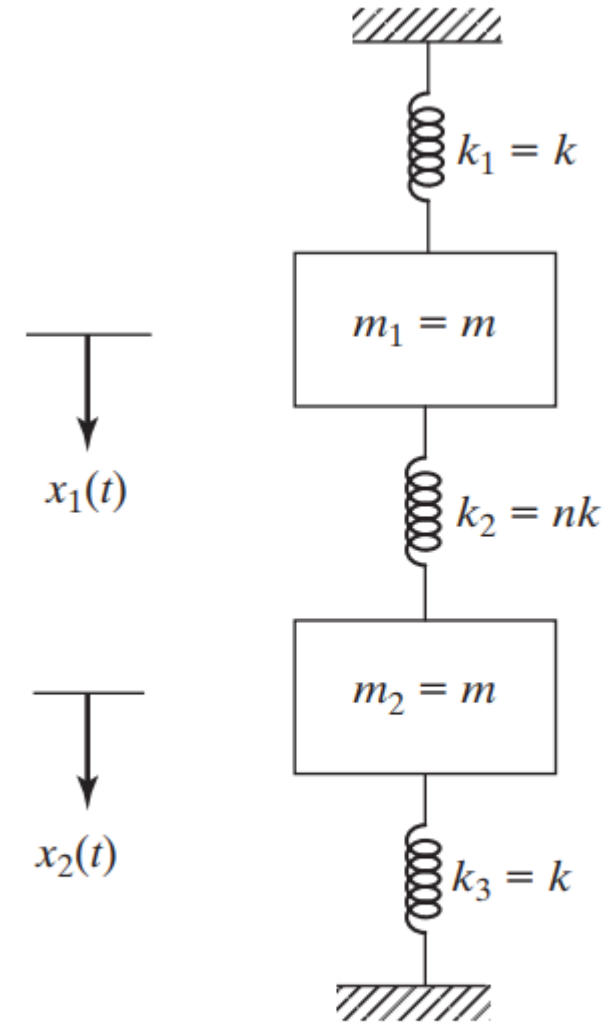


FIGURE 5.6 Two-degree-of-freedom system.

Solution: A 2DOF spring-mass system

Solution: If we measure x_1 and x_2 from the static equilibrium positions of the masses m_1 and m_2 , respectively, the equations of motion and the solution obtained for the system of Fig. 5.5(a) are also applicable to this case if we substitute $m_1 = m_2 = m$ and $k_1 = k_2 = k_3 = k$. Thus the equations of motion, are given by

$$\begin{aligned} m\ddot{x}_1 + 2kx_1 - kx_2 &= 0 \\ m\ddot{x}_2 - kx_1 + 2kx_2 &= 0 \end{aligned} \quad (\text{E.1})$$

By assuming harmonic solution as

$$x_i(t) = X_i \cos(\omega t + \phi); i = 1, 2 \quad (\text{E.2})$$

the frequency equation can be obtained by substituting Eq. (E.2) into Eq. (E.1):

$$\begin{vmatrix} (-m\omega^2 + 2k) & (-k) \\ (-k) & (-m\omega^2 + 2k) \end{vmatrix} = 0$$

or

$$m^2\omega^4 - 4km\omega^2 + 3k^2 = 0 \quad (\text{E.3})$$

Solution: A 2DOF spring-mass system

The solution of Eq. (E.3) gives the natural frequencies

$$\omega_1 = \left\{ \frac{4km - [16k^2m^2 - 12m^2k^2]^{1/2}}{2m^2} \right\}^{1/2} = \sqrt{\frac{k}{m}} \quad (\text{E.4})$$

$$\omega_2 = \left\{ \frac{4km + [16k^2m^2 - 12m^2k^2]^{1/2}}{2m^2} \right\}^{1/2} = \sqrt{\frac{3k}{m}} \quad (\text{E.5})$$

the amplitude ratios are given by

$$r_1 = \frac{X_2^{(1)}}{X_1^{(1)}} = \frac{-m\omega_1^2 + 2k}{k} = \frac{k}{-m\omega_1^2 + 2k} = 1 \quad (\text{E.6})$$

$$r_2 = \frac{X_2^{(2)}}{X_1^{(2)}} = \frac{-m\omega_2^2 + 2k}{k} = \frac{k}{-m\omega_2^2 + 2k} = -1 \quad (\text{E.7})$$

Solution: A 2DOF spring-mass system

The natural modes are given

$$\text{First mode} = \vec{x}^{(1)}(t) = \begin{Bmatrix} X_1^{(1)} \cos\left(\sqrt{\frac{k}{m}}t + \phi_1\right) \\ X_1^{(1)} \cos\left(\sqrt{\frac{k}{m}}t + \phi_1\right) \end{Bmatrix} \quad (\text{E.8})$$

$$\text{Second mode} = \vec{x}^{(2)}(t) = \begin{Bmatrix} X_1^{(2)} \cos\left(\sqrt{\frac{3k}{m}}t + \phi_2\right) \\ -X_1^{(2)} \cos\left(\sqrt{\frac{3k}{m}}t + \phi_2\right) \end{Bmatrix} \quad (\text{E.9})$$

where the constants $X_1^{(1)}$, $X_1^{(2)}$, ϕ_1 , and ϕ_2 are determined by the initial conditions.

Solution: A 2DOF spring-mass system

It can be seen from Eq. (E.8) that when the system vibrates in its first mode, the amplitudes of the two masses remain the same. This implies that the length of the middle spring remains constant. Thus the motions of m_1 and m_2 are in phase (see Fig. 5.7(a)). When the system vibrates in its second mode, Eq. (E.9) shows that the displacements of the two masses have the same magnitude with opposite signs. Thus the motions of m_1 and m_2 are 180° out of phase (see Fig. 5.7(b)). In this case the midpoint of the middle spring remains stationary for all time t . Such a point is called a *node*.

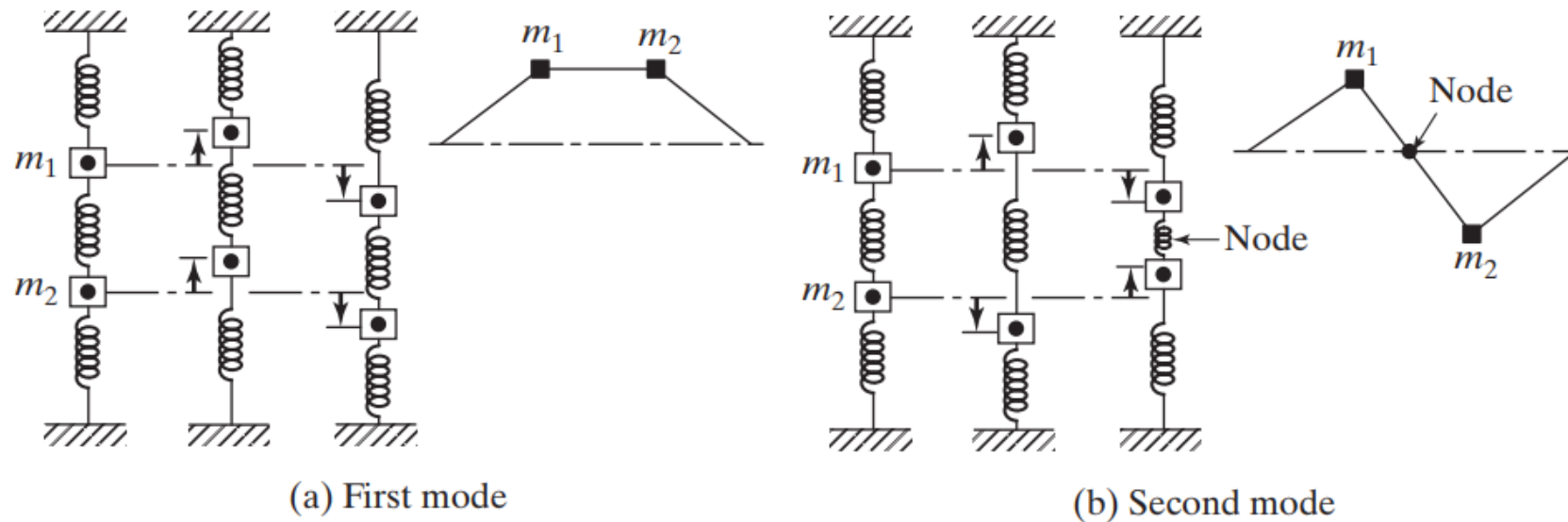


FIGURE 5.7 Modes of vibration.

Solution: A 2DOF spring-mass system

The motion (general solution) of the system can be expressed as

$$\begin{aligned}x_1(t) &= X_1^{(1)} \cos\left(\sqrt{\frac{k}{m}}t + \phi_1\right) + X_1^{(2)} \cos\left(\sqrt{\frac{3k}{m}}t + \phi_2\right) \\x_2(t) &= X_1^{(1)} \cos\left(\sqrt{\frac{k}{m}}t + \phi_1\right) - X_1^{(2)} \cos\left(\sqrt{\frac{3k}{m}}t + \phi_2\right)\end{aligned}\tag{E.10}$$

Note: It can be seen that the computation of the natural frequencies and mode shapes is lengthy and tedious. Computer programs can be used conveniently for the numerical computation of the natural frequencies and mode shapes of multi-degree-of-freedom systems

Example: A 2DOF spring-mass-damper system

Find the free-vibration response of the system shown in Fig. 5.5(a) with $k_1 = 30$, $k_2 = 5$, $k_3 = 0$, $m_1 = 10$, $m_2 = 1$, and $c_1 = c_2 = c_3 = 0$ for the initial conditions $x_1(0) = 1$, $\dot{x}_1(0) = \dot{x}_2(0) = \dot{x}_2(0) = 0$.

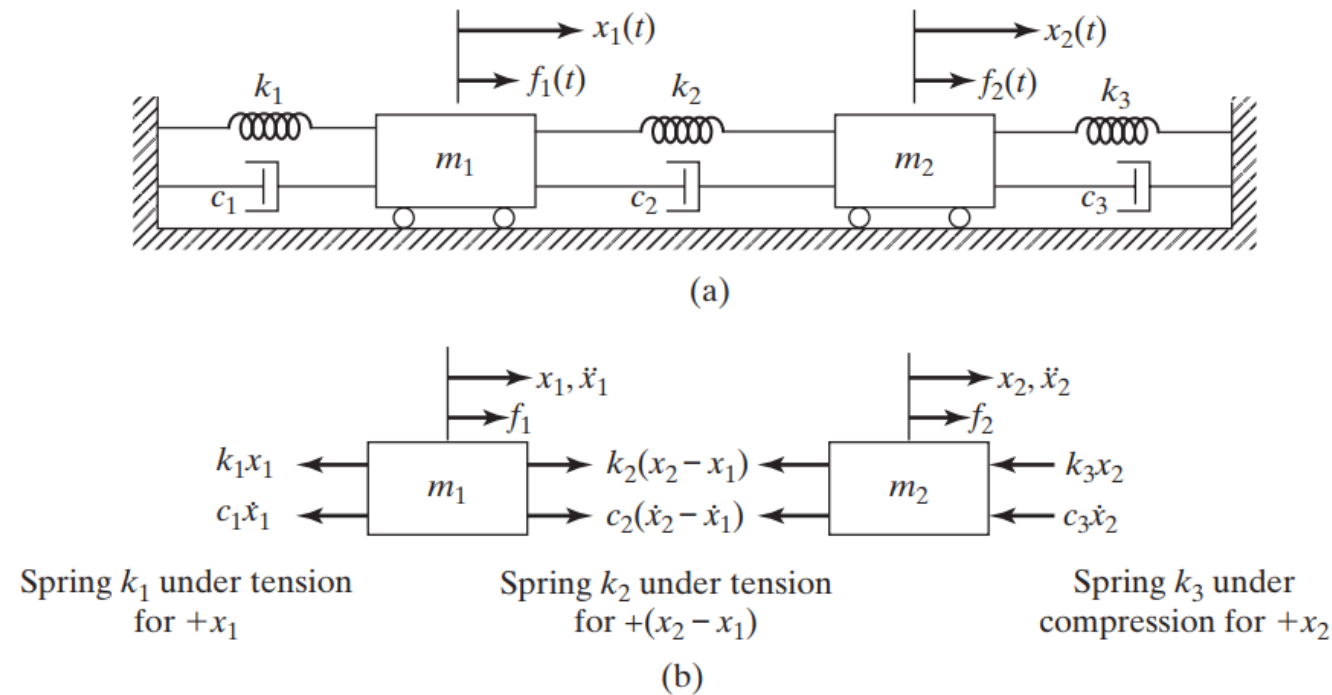


FIGURE 5.5 A two-degree-of-freedom spring-mass-damper system.

Solution: A 2DOF spring-mass-damper system

Solution: For the given data, the eigenvalue problem, becomes

$$\begin{bmatrix} -m_1\omega^2 + k_1 + k_2 & -k_2 \\ -k_2 & -m_2\omega^2 + k_2 + k_3 \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

or

$$\begin{bmatrix} -10\omega^2 + 35 & -5 \\ -5 & -\omega^2 + 5 \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad (\text{E.1})$$

By setting the determinant of the coefficient matrix in Eq. (E.1) to zero, we obtain the frequency equation

$$10\omega^4 - 85\omega^2 + 150 = 0 \quad (\text{E.2})$$

from which the natural frequencies can be found as

$$\omega_1^2 = 2.5, \quad \omega_2^2 = 6.0$$

or

$$\omega_1 = 1.5811, \quad \omega_2 = 2.4495 \quad (\text{E.3})$$

Solution: A 2DOF spring-mass-damper system

The substitution of $\omega^2 = \omega_1^2 = 2.5$ in Eq. (E.1) leads to $X_2^{(1)} = 2X_1^{(1)}$, while $\omega^2 = \omega_2^2 = 6.0$ in Eq. (E.1) yields $X_2^{(2)} = -5X_1^{(2)}$. Thus the normal modes (or eigenvectors) are given by

$$\vec{X}^{(1)} = \begin{Bmatrix} X_1^{(1)} \\ X_2^{(1)} \end{Bmatrix} = \begin{Bmatrix} 1 \\ 2 \end{Bmatrix} X_1^{(1)} \quad (\text{E.4})$$

$$\vec{X}^{(2)} = \begin{Bmatrix} X_1^{(2)} \\ X_2^{(2)} \end{Bmatrix} = \begin{Bmatrix} 1 \\ -5 \end{Bmatrix} X_1^{(2)} \quad (\text{E.5})$$

The free-vibration responses of the masses m_1 and m_2 are given by

$$x_1(t) = X_1^{(1)} \cos(1.5811t + \phi_1) + X_1^{(2)} \cos(2.4495t + \phi_2) \quad (\text{E.6})$$

$$x_2(t) = 2X_1^{(1)} \cos(1.5811t + \phi_1) - 5X_1^{(2)} \cos(2.4495t + \phi_2) \quad (\text{E.7})$$

where $X_1^{(1)}$, $X_1^{(2)}$, ϕ_1 , and ϕ_2 are constants to be determined from the initial conditions. By using the given initial conditions in Eqs. (E.6) and (E.7), we obtain

$$x_1(t=0) = 1 = X_1^{(1)} \cos \phi_1 + X_1^{(2)} \cos \phi_2 \quad (\text{E.8})$$

$$x_2(t=0) = 0 = 2X_1^{(1)} \cos \phi_1 - 5X_1^{(2)} \cos \phi_2 \quad (\text{E.9})$$

$$\dot{x}_1(t=0) = 0 = -1.5811X_1^{(1)} \sin \phi_1 - 2.4495X_1^{(2)} \sin \phi_2 \quad (\text{E.10})$$

$$\dot{x}_2(t=0) = -3.1622X_1^{(1)} \sin \phi_1 + 12.2475X_1^{(2)} \sin \phi_2 \quad (\text{E.11})$$

Solution: A 2DOF spring-mass-damper system

The solution of Eqs. (E.8) and (E.9) yields

$$X_1^{(1)} \cos \phi_1 = \frac{5}{7}, \quad X_1^{(2)} \cos \phi_2 = \frac{2}{7} \quad (\text{E.12})$$

while the solution of Eqs. (E.10) and (E.11) leads to

$$X_1^{(1)} \sin \phi_1 = 0, \quad X_1^{(2)} \sin \phi_2 = 0 \quad (\text{E.13})$$

Equations (E.12) and (E.13) give

$$X_1^{(1)} = \frac{5}{7}, \quad X_1^{(2)} = \frac{2}{7}, \quad \phi_1 = 0, \quad \phi_2 = 0 \quad (\text{E.14})$$

Thus the free-vibration responses of m_1 and m_2 are given by

$$x_1(t) = \frac{5}{7} \cos 1.5811t + \frac{2}{7} \cos 2.4495t \quad (\text{E.15})$$

$$x_2(t) = \frac{10}{7} \cos 1.5811t - \frac{10}{7} \cos 2.4495t \quad (\text{E.16})$$

Example: A half-car model traversing a road profile

Derive the equations of motion of the system shown in Fig. 5.20.

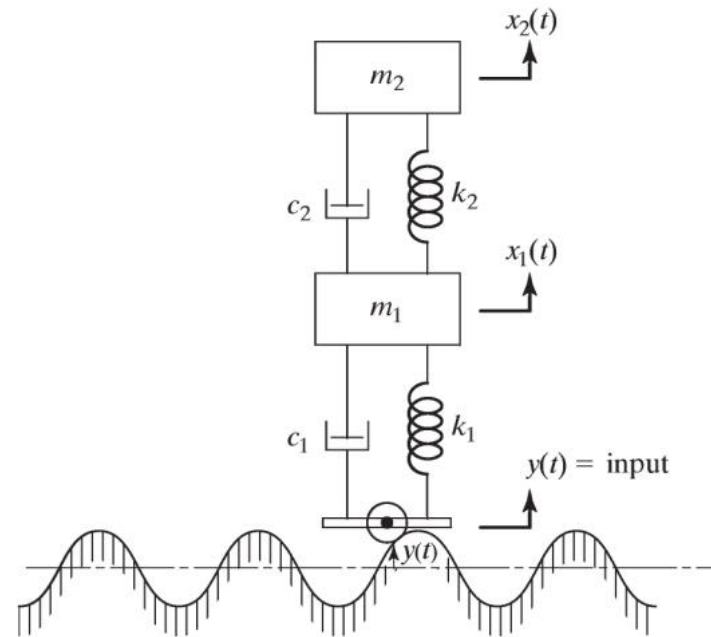


FIGURE 5.20

Solution: A half-car model traversing a road profile

Solution:

First draw the free body diagrams of masses m_1 and m_2 :

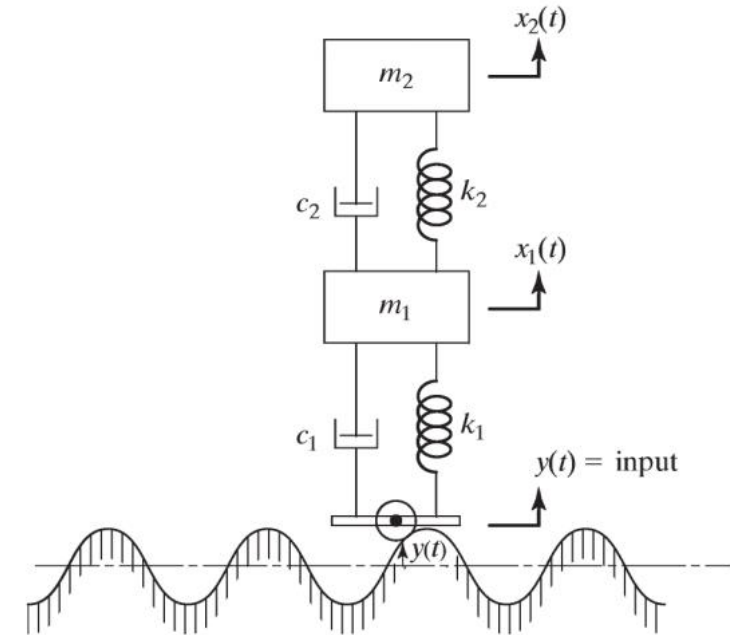
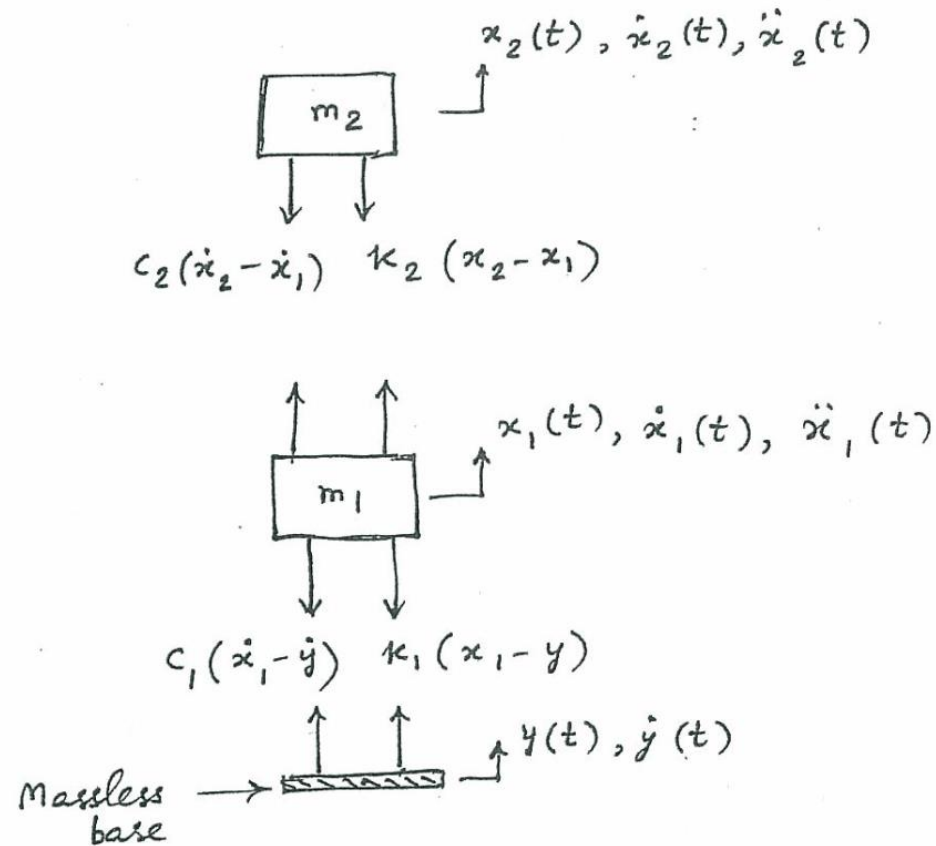


FIGURE 5.20

Solution: A half-car model traversing a road profile

Equations of motion (using Newton's second law of motion)

For mass m_2 :

$$m_2 \ddot{x}_2 + c_2(\dot{x}_2 - \dot{x}_1) + k_2(x_2 - x_1) = 0 \quad (1)$$

For mass m_1 :

$$m_1 \ddot{x}_1 + c_1(\dot{x}_1 - \dot{y}) + k_1(x_1 - y) - c_2(\dot{x}_2 - \dot{x}_1) - k_2(x_2 - x_1) = 0 \quad (2)$$

Eq. (1) and (2) can be rewritten as:

$$m_1 \ddot{x}_1 + (c_1 + c_2)\dot{x}_1 - c_2\dot{x}_2 + (k_1 + k_2)x_1 - k_2x_2 = c_1\dot{y} + k_1y \quad (3)$$

$$m_2 \ddot{x}_2 + c_2\dot{x}_2 - c_2\dot{x}_1 + k_2x_2 - k_2x_1 = 0 \quad (4)$$

Eq. (3) and (4) can be expressed in matrix form as:

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} c_1 + c_2 & -c_2 \\ -c_2 & c_2 \end{bmatrix} \begin{Bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{Bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} c_1\dot{y} + k_1y \\ 0 \end{Bmatrix} \quad (5)$$

Example: A 2DOF spring-mass-pendulum

A two-mass system consists of a piston of mass m_1 , connected by two elastic springs, that moves inside a tube as shown in Fig. 5.23. A pendulum of length l and end mass m_2 , is connected to the piston as shown in Fig. 5.23.

- (a) Derive the equations of motion of the system in terms of $x_1(t)$ and $\theta(t)$.
- (b) Derive the equations of motion of the system in terms of the $x_1(t)$ and $x_2(t)$.
- (c) Find the natural frequencies of vibration of the system.

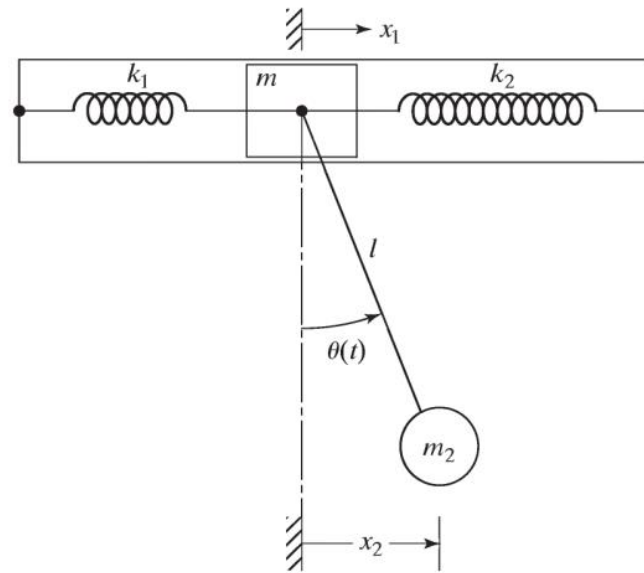
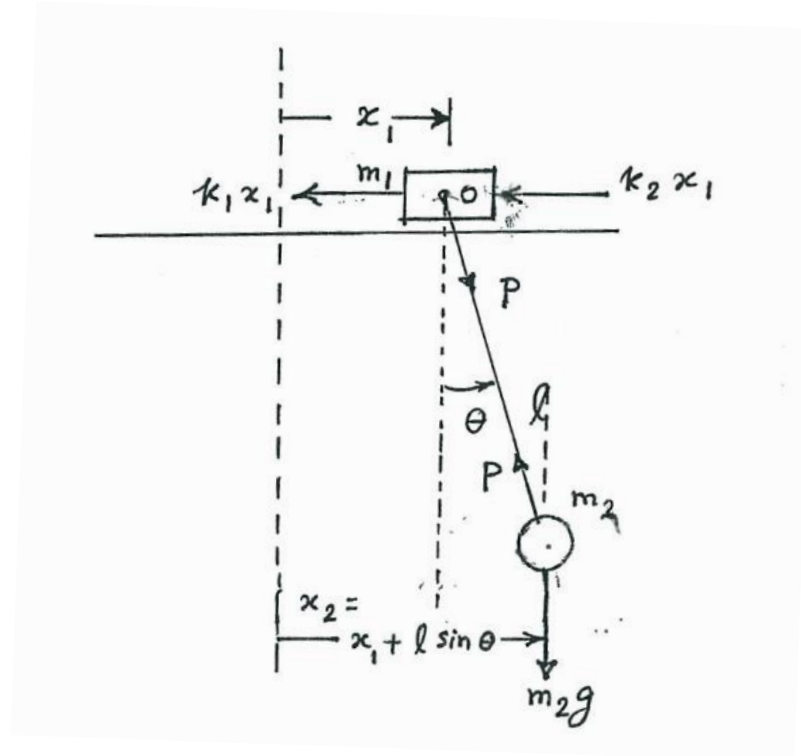


FIGURE 5.23

Example: A 2DOF spring-mass-pendulum

First draw the force analysis diagram:



(a) Using $x_1(t)$ and $\theta(t)$:

Equation of motion of the piston:

$$m_1 \ddot{x}_1 + (k_1 + k_2)x_1 - P \sin \theta = 0 \quad (1)$$

Equation of motion of the pendulum bob:

For the dynamic equilibrium of the bob, take moments about the pivot point:

$$m_2 l^2 \ddot{\theta} - m_2 g l \sin \theta = 0 \quad (2)$$

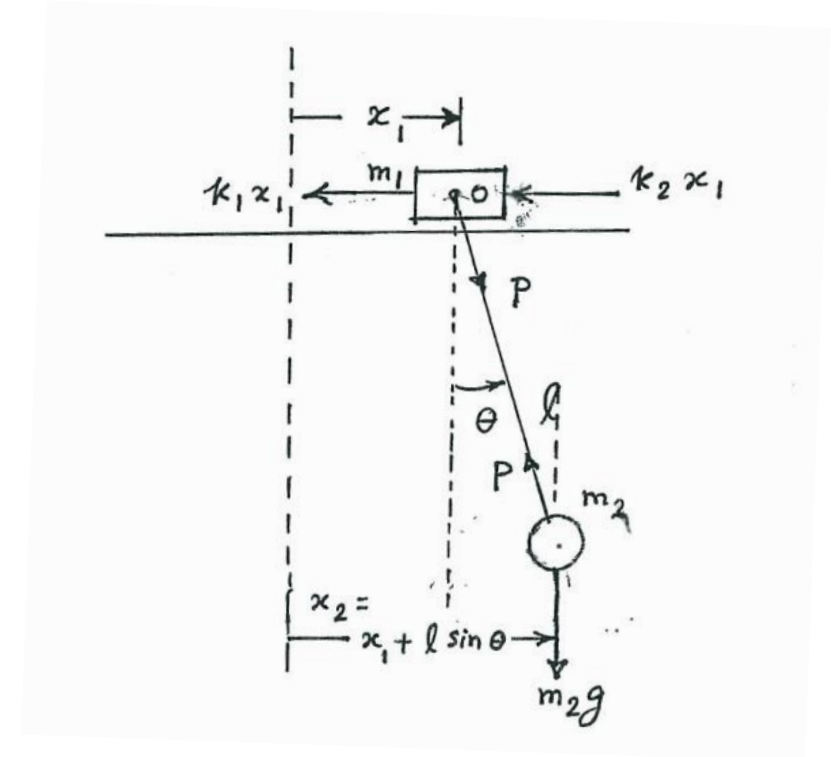
Assuming small angle θ , we can substitute: $\sin \theta \approx \theta$

$$P = m_2 g \cos \theta \approx m_2 g$$

In Eq. (1) and (2) to obtain the equations of motion as:

$$m_1 \ddot{x}_1 + (k_1 + k_2)x_1 - m_2 g \theta = 0 \quad (3)$$

$$m_2 l^2 \ddot{\theta} - m_2 g l \theta = 0 \quad (4)$$



(b) Using $x_1(t)$ and $x_2(t)$:

Equation of motion of the piston:

$$m_1 \ddot{x}_1 + (k_1 + k_2)x_1 - P \sin \theta = 0 \quad (5)$$

Equation of motion of the pendulum bob:

$$m_2 \ddot{x}_2 + P \sin \theta = 0 \quad (6)$$

In Eq. (5) and (6), $\sin \theta$ can be expressed as: $\sin \theta \approx \theta = \frac{x_2 - x_1}{l} \quad (7)$

for small angle θ ,

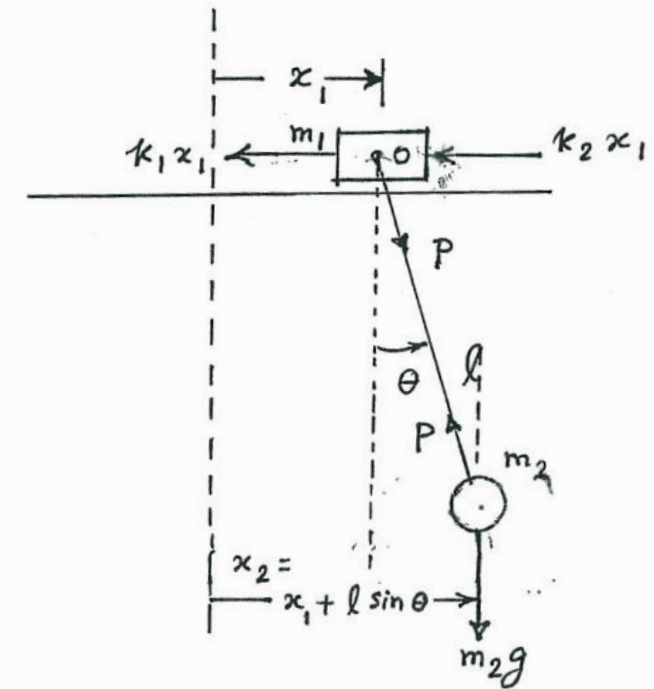
For the vertical equilibrium of mass m_2 ,

$$P = m_2 g \cos \theta \approx m_2 g \quad (8)$$

Substitution of Eq. (7) and (8) in Eq. (5) and (6) gives:

$$m_1 \ddot{x}_1 + (k_1 + k_2)x_1 - m_2 g \left(\frac{x_2 - x_1}{l} \right) = 0 \quad (9)$$

$$m_2 \ddot{x}_2 + m_2 g \left(\frac{x_2 - x_1}{l} \right) = 0 \quad (10)$$



(c) Assuming harmonic solutions as:

$$\begin{cases} x_1(t) = X_1 \cos(\omega t + \phi) \\ x_2(t) = X_2 \cos(\omega t + \phi) \end{cases} \quad (11)$$

Eq. (9) and (10) become:

$$\left\{ -m_1 \omega^2 + (k_1 + k_2) + \frac{m_2 g}{l} \right\} X_1 - \frac{m_2 g}{l} X_2 = 0 \quad (12)$$

$$-\frac{m_2 g}{l} X_1 + \left\{ -m_2 \omega^2 + \frac{m_2 g}{l} \right\} X_2 = 0 \quad (13)$$

By setting the determinant of the coefficient matrix of X_1 and X_2 , we obtain the frequency equation as:

$$\begin{vmatrix} -m_1 \omega^2 & (k_1 + k_2 + \frac{m_2 g}{l}) \\ -\frac{m_2 g}{l} & -m_2 \omega^2 + \frac{m_2 g}{l} \end{vmatrix} = 0 \quad (14)$$

$$m_1 m_2 \omega^4 - \frac{m_1 m_2 g}{l} \omega^2 + \frac{m_2 g}{l} \left(k_1 + k_2 + \frac{m_2 g}{l} \right) = 0 \quad (15)$$

The roots of Eq. (15) give the natural frequency of vibration of the system.

Example: Double pendulum

Set up the differential equations of motion for the double pendulum shown in Fig. 5.25, using the coordinates x_1 and x_2 and assuming small amplitudes. Find the natural frequencies, the ratios of amplitudes, and the locations of nodes for the two modes of vibration when $m_1 = m_2 = m$ and $\ell_1 = \ell_2 = \ell$.

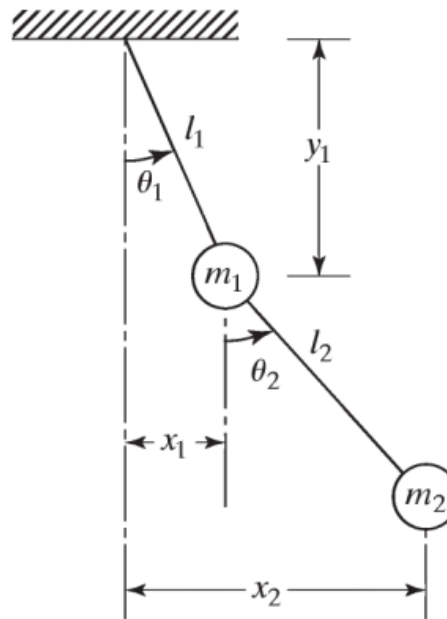


FIGURE 5.25

Solution for Example: Double pendulum

Taking moment about O and equation for mass m_1 is:

$$m_1 \ell_1^2 \ddot{\theta}_1 = -w_1(\ell_1 \sin \theta_1) + Q \sin \theta_2 (\ell_1 \cos \theta_1) - Q \cos \theta_2 (\ell_1 \sin \theta_1)$$

Assuming small amplitudes of vibration and $Q \approx w_2$, the above equation can be simplified to:

$$m_1 \ell_1^2 \ddot{\theta}_1 = -w_1 \ell_1 \theta_1 + w_2 \ell_1 (\theta_2 - \theta_1) \quad (E_1)$$

Similarly, one can obtain the equation of motion for mass m_2

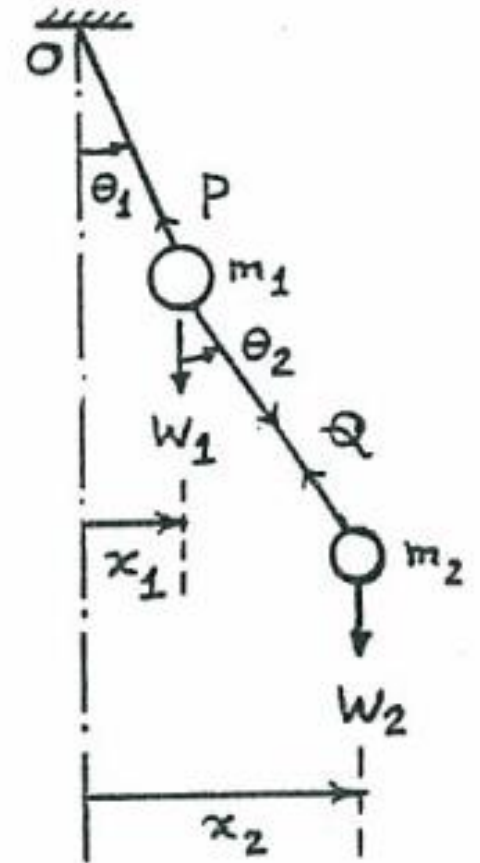
$$m_2 \ell_2^2 \ddot{\theta}_2 + m_2 \ell_2 (\ell_1 \ddot{\theta}_1) = -w_2 (\ell_2 \sin \theta_2) = -w_2 \ell_2 \theta_2 \quad (E_2)$$

Employing the relations $\theta_1 = \frac{x_1}{\ell_1}$ and $\theta_2 = \frac{x_2 - x_1}{\ell_2}$, Eqs. (E₁) and (E₂) become

$$m_1 \ell_1 \ddot{x}_1 + \left[w_1 + w_2 \left(\frac{\ell_1 + \ell_2}{\ell_2} \right) \right] x_1 - \frac{w_2 \ell_1}{\ell_2} x_2 = 0 \quad (E_3)$$

$$m_2 \ell_2 \ddot{x}_2 - w_2 x_1 + w_2 x_2 = 0 \quad (E_4)$$

Note that eqns can also be obtained by energy method, though more difficult using the coordinates of x_1 and x_2 .



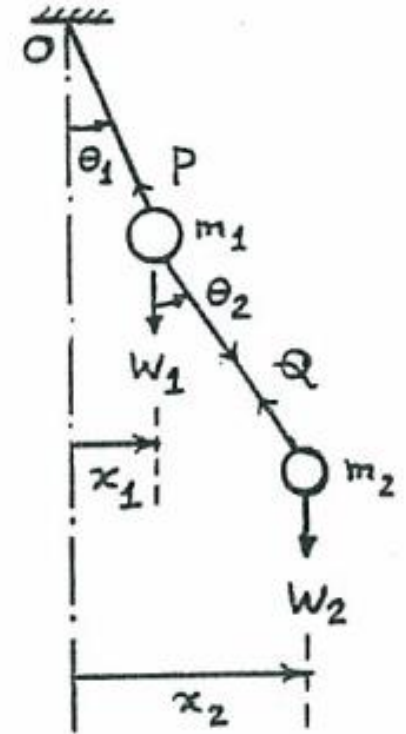
For $m_1 = m_2 = m$, $\ell_1 = \ell_2 = \ell$, we have $w_1 = w_2 = mg$, then Eqs. (E₁) and (E₂) are simplified to:

$$\begin{aligned} m\ell\ddot{x}_1 + 3mgx_1 - mgx_2 &= 0 \\ m\ell\ddot{x}_2 - mgx_1 + mgx_2 &= 0 \end{aligned} \quad (\text{E}_5)$$

Assume harmonic motions $x_i(t) = X_i \cos \omega t$, $i = 1, 2$, for the free vibration, Eq. (E₅) becomes

$$\begin{aligned} -\omega^2 m\ell X_1 + 3mgX_1 - mgX_2 &= 0 \\ -\omega^2 m\ell X_2 - mgX_1 + mgX_2 &= 0 \end{aligned} \quad (\text{E}_6)$$

which is of the form $\mathbf{A}\mathbf{X} = 0$. The vanishing of the determinant of $\mathbf{A}$ gives the frequency equation. Note that Eq. (E₅) can also be cast in the standard matrix form of $\mathbf{M}\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = 0$, leading to the eigenvalue problem of $\mathbf{K}\mathbf{u} = \omega^2 \mathbf{M}\mathbf{u}$. The vanishing of the determinant of $(\mathbf{K} - \omega^2 \mathbf{M})$ gives the frequency equation.



Evaluating $\det(\mathbf{K} - \omega^2 \mathbf{M}) = 0$ gives the frequency equation:

$$\omega^4 m^2 \ell^2 - (4m^2 \ell g) \omega^2 + 2m^2 g^2 = 0, \text{ i.e., } \omega_1^2, \omega_2^2 = (2 \pm \sqrt{2}) \frac{g}{\ell}$$

The natural frequencies can be evaluated as:

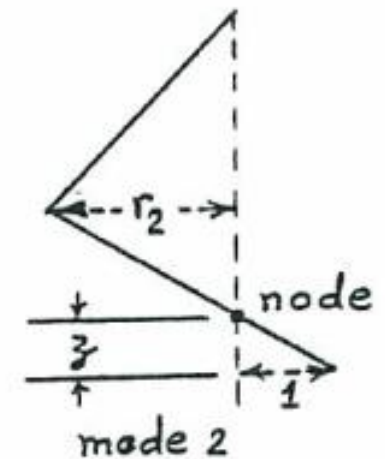
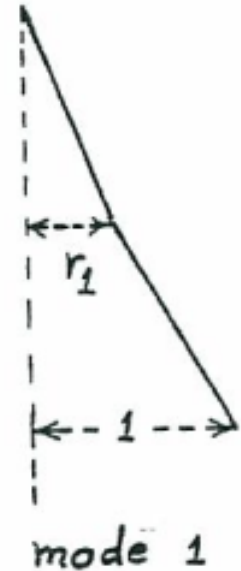
$$\omega_1 = 0.7654 \sqrt{\frac{g}{\ell}}, \quad \omega_2 = 1.8478 \sqrt{\frac{g}{\ell}}$$

Substituting these values into the eigenvalue problem, Eq. (E₆), can give the amplitude ratios for each mode.

$$\frac{X_1}{X_2} = \frac{mg}{-\omega^2 m \ell + 3mg} = \frac{1}{-(\omega^2 \ell)/g + 3}$$

For mode 1, $\omega_1 = 0.7654 \sqrt{\frac{g}{\ell}}, r_1 = \left(\frac{X_1}{X_2}\right)^{(1)} = 0.4142$. No node

For mode 2, $\omega_2 = 1.8478 \sqrt{\frac{g}{\ell}}, r_1 = \left(\frac{X_1}{X_2}\right)^{(2)} = -2.4133$. Employing similar triangle principle, location of the node is at z : $\frac{z}{1} = \frac{1-z}{2.4133}$ or $z = 0.2930$.



Example: Two masses on a string

Figure 5.33 shows a system of two masses attached to a tightly stretched string, fixed at both ends. Determine the natural frequencies and mode shapes of the system for $m_1 = m_2 = m$ and $\ell_1 = \ell_2 = \ell_3 = \ell$.

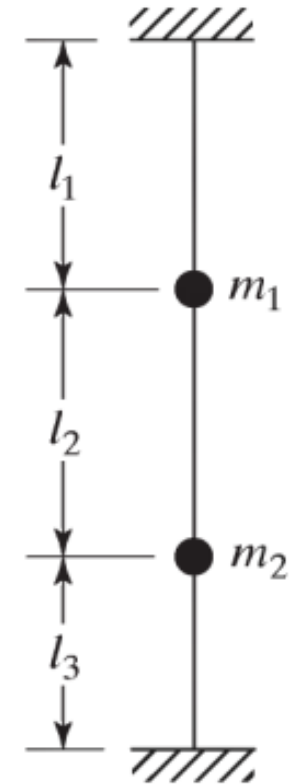


FIGURE 5.33

Solution: Two masses on a string

Let P be the tension in the string. Horizontal component of tension (along x_1 direction) in the string lying above and below m_1 are:

$$-\frac{x_1 P}{\ell_1} \text{ and } -\frac{(x_1 - x_2)P}{\ell_2}, \text{ respectively}$$

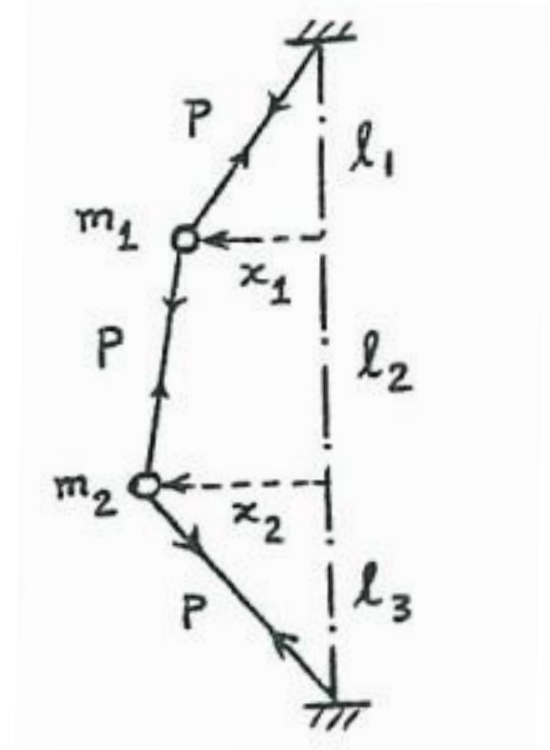
Treat the system in the horizontal plane as we have learned that gravity force does not affect the characteristics of linear vibratory systems.

Applying Newton's second law gives:

$$m_1 \ddot{x}_1 = -\frac{x_1 P}{\ell_1} - \frac{(x_1 - x_2)P}{\ell_2} \quad \text{or} \quad m_1 \ddot{x}_1 + \frac{x_1 P}{\ell_1} + \frac{(x_1 - x_2)P}{\ell_2} = 0$$

Similarly, by symmetry, we can obtain governing equation for m_2 :

$$m_2 \ddot{x}_2 + \frac{x_2 P}{\ell_3} + \frac{(x_2 - x_1)P}{\ell_2} = 0$$



Eigenvalue problem for the free vibration analysis:

$$\begin{bmatrix} \left(-m\omega^2 + \frac{2P}{\ell}\right) & -\frac{P}{\ell} \\ -\frac{P}{\ell} & \left(-m\omega^2 + \frac{2P}{\ell}\right) \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad (\text{E1})$$

Vanishing of the determinant of the above matrix gives the frequency equation:

$$\left(-m\omega^2 + \frac{2P}{\ell}\right)^2 - \left(\frac{P}{\ell}\right)^2 = \left(-m\omega^2 + \frac{3P}{\ell}\right)\left(-m\omega^2 + \frac{P}{\ell}\right) = 0 \Rightarrow \omega_1 = \sqrt{\frac{P}{m\ell}}, \omega_2 = \sqrt{\frac{3P}{m\ell}}$$

Using the first of Eq. (E1), we can find the ratio of the amplitude of the two physical coordinates,

$$r_1 = \frac{X_2^{(1)}}{X_1^{(1)}} = \frac{-m\omega_1^2 + \frac{2P}{\ell}}{\left(\frac{P}{\ell}\right)} = 1, \quad r_2 = \frac{X_2^{(2)}}{X_1^{(2)}} = \frac{-m\omega_2^2 + \frac{2P}{\ell}}{\left(\frac{P}{\ell}\right)} = -1$$

The mode shapes are thus:

$$\vec{X}^{(1)} = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} X_1^{(1)} \text{ (no nodal point), } \vec{X}^{(2)} = \begin{Bmatrix} 1 \\ -1 \end{Bmatrix} X_1^{(2)} \text{ (one nodal point at the middle of the two masses)}$$